Psychology 1 1 1 1 - Take note on the main points

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Topic One: Introduction to Human psychology.

Topic Objectives

By end of this topic, you should be able to:-

Describe psychology, historical background, goals of psychology and schools of thought

Definition

- Ø The word psychology comes from the roots ‘psyche’ and ‘logos’, which means knowledge and study respectively.
- Ø Psychology is the scientific study of mental processes, behavior and the relationship between them.
- Ø It’s the social science of the human mind and it’s interaction with the environment.
- Ø It’s not an absolute science because it deals with human thoughts, feelings and behavior and as we are aware, humans are not always predictable and reliable.
- Ø Behavior means all that people do. Considers physical activities, attitudes and emotional changes.
- We study behavior when we want to understand why the particular behavior occurs
-
- People behave the way they do for various reasons. It may be learning, physical problems, or psychological problems
-
- **Historical background**
- Psychology began as philosophy which entails attempting to understand behavior through reasoning.
- Philosophy is defined as the study of the mind.
- Philosophy did not involve observations or experiments.
- Behavior was explained through reasoning and arguments.
- Socrates is known for questioning everything and supplying no answers.
- Plato’s student attempted to explain why people dream and how people operate
- The Greeks were the first to define psychology as the study of the mind or the soul.

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- Aristotle, a Greek philosopher thought that the highest form of reasoning and thinking came from the heart.
- But during the 19th century Psychology became a science and an academic discipline when people who wanted to learn more about human behavior and mental processes
- During this time a German scientist Wilhelm Wundt opened the first laboratory and faculty of Psychology in the University of Leipzig in 1879 and established the first Psychological Journal in 1881.
- He was the first to apply the scientific method to the field.
- His work focused on consciousness – or the awareness of immediate experience.
- However, it applied the efforts of the scientific process in the same way physics or chemistry would.
- Later, one of his students G. Stanley Hall brought the science of psychology to America.
- He founded the American Psychological Association and became its first president.
- He started America's first research laboratory at John's Hopkins University in 1883, and published the first journal of psychology in America.

Goals of psychology

- Goals of psychology
- Describe **– what is the nature of this behavior?**
- **The first goal is to observe behavior and describe, often in as much detail as possible, what was observed as objectively as possible.**
- **Through describing the behavior of humans and other animals, we are able to understand it and gain a better perspective on what is considered normal and abnormal.**
- Explain **–why does it occur?**
- **While descriptions come from observable data, psychologists must go beyond what is obvious**

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and explain their observations. In other words, why did the subject do what he or she did?

- Predict **–**
- **Another primary goal of psychology is to make predictions.**
- **Once we know what happens, and why it happens, we can begin to speculate what will happen in the future.**
- **There's an old saying, which very often holds true: "the best predictor of future behavior is past behavior."**
- Control **–The fourth goal of psychology is to control, modify or change the existing behavior. Influencing behavior in a helpful manner.**
- **Once we know what happens, why it happens and what is likely to happen in the future, we can expect control over it.**
- **E.g, if your father was abusive, you are likely to choose an abusive partner, and can therefore intervene to change this negative behavior**
- Improve –**Not only do psychologists attempt to control behavior, they want to do so in a positive manner, they want to improve a person's life, not make it worse. This is not always the case, but it should always be the intention.**
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Schools of thought

Schools of thought

- General points of view used in explaining people's behavior and thinking, whether normal or abnormal.
- Each viewpoint approaches psychology from a slightly different perspective.

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- Ø A historical /Early schools of thought
- Ø Structuralism
- Ø Functionalism
- Ø Modern approaches /perspectives
- Ø Behaviorism
- Ø psychoanalytic
- Ø Humanistic
- Ø Cognitive
- Ø biological
- Ø evolution
- Ø social cultural
- **Early schools of thought**
- **Structuralism**
- The first formal school of thought in Psychology,
- Structuralists were led by Edward Titchner, one of Wundt students who believed that the task of psychology was to analyze consciousness into its basic elements and study how these were related. These elements would include ideas like sensations, emotions, and images.
- **Functionalism**; (why do we do what we do)
- An early school of thought concerned with how humans and animals use mental processes in adapting to their environment.
- It was formed as a reaction to the theories of structuralism school of thought.
- William James, one of the founders of functionalism, set out to explain <u>why we think the way we do</u>.
- He felt that psychology should have practical values and that psychologists should find out how the mind can function to a person's benefit.
-

Topic One: Summary

Psychology is the scientific study of mental processes, behavior and the relationship between them. Its main goals include; describe, explain, predict, control and improve. it entails traditional schools of thought e.g structuralism and functionalism and modern schools of thought e,g. behavior, psychoanalytic, humanistic, evolution, cognitive and social cultural.

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Topic Two: Modern approaches to psychology

Modern approach /perspective

- Behavioral; An approach to psychology emphasizing the scientific study of observable behavioral responses and their environmental determinants
- Proponents

- Ivan Pavlov (behaviorist) Started the idea of Classical conditioning,
- John Watson (1878-1958).
- B. .F Skinner (1904-1990). (operant conditioning and reinforcement)
- **Psychodynamic**; Emphasizes unconscious thought, the conflict between biological drives (such as the drive to serve) and society's demands and early childhood family experiences.
- Proponent :Sigmund Freud ,Erickson Neo Freudian
- **Humanistic;** Emphasizes a person's positive qualities, their capacity for growth, and the freedom to choose their destiny.
- Proponent: Carl Rogers and Abraham Maslow
- **Cognitive;** emphasizes the mental processes involved in knowing, how we direct our attention, perceive, remember, think and solve problems
- **Biological;** Focuses on the body especially the brain and the nervous system, the functioning of the neurons, neurotransmitters and hormones, and heredity, to explain behavior.
- **Evolutionary Psychology :** centered on evolutionary ideas such as adaptation, reproduction and natural selection as the basis for explaining specific human behaviors
- **Social-Cultural;** examines the ways in which social and cultural environments influence behavior.
- stresses the importance of understanding those influences when interpretating the behaviors of others
- Cultural relativity promotes the view that different cultures, ethnic groups, and genders, are simply different from, rather than inferior to, others
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Methods used for studying psychology

- **Method used in studying psychology**
- Methods of Studying Human Behavior are as follows:
- **Experimental Method:**
- It's the most objective way of studying behavior here, experiments are conducted in the laboratories under controlled conditions.
- **Case studies and Clinical Studies**
- Case studies allow researchers to study a single individual or group of people in great depth.
- The researcher collects every single piece of data possible about the subject, including
- Hospital, educational, family background and all other necessary data. often observing the person of interest over a period and in a variety of situations
- **Survey Method:**
- This is used to gather the information from large number of people. Questionnaires, checklists, rating scales, inventories are used to collect the required information
- This method is usually used to gather information about political opinion, customers' preferences, etc.
- **Observation:** Two kinds of Observation method
- *Participant*
- • The researcher has to join the group under study in order to gather deeper and more intensive data.
- *Non-Participant / Naturalistic*
- • The researcher is just a plain observer. Oftentimes, the subjects do not know that they are under study.
- **Test Method**
- uses Psychological Test as its instrument.
- Examples of Psychological Test are:
- - IQ Test
- -Problem Checklist
- -Aptitude Test
- Free Association Technique -> used to study patients with mental problems
- -Study Habit Checklist

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Branches of studying psychology

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- **Branches of psychology**
- There are many specialties within psychology, this includes:
- **Clinical Psychology**: specialize in the diagnosis and treatment of mental and behavioral disorders, such as anxiety, phobias, and schizophrenia.
- **Developmental Psychology:** study how people grow, develop, and change throughout the life span
- **Educational Psychology:** The study of how humans learn in educational settings, the effectiveness of educational interventions, the psychology of teaching.
- **Social Psychology:** investigate how the individual feels, thinks, and behaves in a social setting—in the presence of others.
- **Industrial/ Organizational Psychology:** study the relationships between people
- and their work environments.
- **Biopsychology**: study the relationship between physiological processes and behavior.
- **Counseling Psychology:** is a professional psychology field that pays attention to the “emotional, social, vocational, educational, health-related, developmental, and organizational concerns” that normally impact our daily lives across lifespan.
- **Experimental Psychology:** conduct experiment in most areas of psychology—
- Learning, memory, sensation, perception, motivation, emotion, and others.
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Topic Two: SummaryFF

The modern approaches to psychology are

Ø Behaviorism

Ø psychoanalytic

Ø Humanistic

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- Cognitive
- biological

branches of psychology include

social, educational, clinical, industrial etc

Methods for studying psychology include, case studies, experimental. Observational and survey

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Unit Two Content..

Topic 1: Theories of Human growth and development

At the end of this topic , the learner shall be able to:

Describe the following theories;

1. Erickson psychosocial theory of development.
2. Attachment theories by John Bowlby and Mary Ainsworth.
3. Moral development theory by Kohlberg.
4. Gilligan's stages of moral development

Topic 1: Theories of Human growth and development(Cont)

Theories of Human growth and development

Theories of development provide a framework for thinking about human growth, development, and learning. Several famous psychologists, including Erickson, Bowlby, Ainswothy, Kolhberg, Gilligans among others describes the human growth and development.Erickson, Bowlby, Ainswothy, Kolhberg, Gilligans have

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developed theories pertaining to human growth and development.These theories will be discussed in this unit.

Erik Erickson theory of psychosocial development.

Erik Erikson believed in the importance of early childhood. However, Erikson believed that personality development happens over the entire course of a person's life.In the early 1960s, Erikson proposed a theory that describes eight distinct stages of development.According to Erikson, in each stage people face new challenges and the stage's outcome depends on how people handle these challenges. These challenges are commonly referred to as crisis.Erikson named the stages according to these possible outcomes:

Stage 1: Trust vs. Mistrust

In the first year after birth, babies depend completely on adults for basic needs such as food, comfort, and warmth. If the caretakers meet these needs reliably, the babies become attached and develop a sense of security. Otherwise, they may
develop a mistrustful, insecure attitude.

Stage 2: Autonomy vs. Shame and Doubt

Between the ages of one and three, toddlers start to gain independence and learn skills such as toilet training, feeding themselves, and dressing themselves. Depending on how they face these challenges, toddlers can develop a sense of autonomy or a sense of doubt and shame about themselves.

Stage 3: Initiative vs. Guilt

Between the ages of three and six, children must learn to control their impulses and act in a socially responsible way.
If they can do this effectively, children become more self- confident. If not, they may develop a strong sense of guilt.

Stage 4: Industry vs. Inferiority

Between the ages of six and twelve, children compete with peers in school and prepare to take on adult roles.
They end this stage with either a sense of competence or a sense of inferiority.

Stage 5: Identity vs. Role Confusion

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- During adolescence, which is the period between puberty and adulthood, children try to determine their identity and their direction in life.

- Depending on their success, they either acquire a sense of identity or remain uncertain about their roles in life.

Stage 6: Intimacy vs. Isolation

In young adulthood, people face the challenge of developing intimate relationships with others. If they do not succeed, they may become isolated and lonely.

- **Stage 7: Generativity vs. Self-Absorption**
- • As people reach middle adulthood, they work to become productive members of society, either through parenting or through their jobs.
- • If they fail, they become overly self-absorbed
- **Stage 8: Integrity vs. Despair**
- • In old age, people examine their lives. They may either have a sense of contentment or be disappointed about their lives and fearful of the future.
- **Summary of Erikson's Theory of psychosocial development**

Stage	Psychosocial Crisis	Basic Virtue	Age
1	Trust vs. mistrust	Hope	Infancy (o to1 ½)
2	Autonomy vs. shame	Will	Early Childhood (1 ½ to3)
3	Initiative vs. guilt	Purpose	Play Age (3 to 5)
4	Industry vs. inferiority	Competency	School Age (5 to 12)
5	Ego identity vs. Role Confusion	Fidelity	Adolescence (12 to 18)
6	Intimacy vs. isolation	Love	Young Adult (18 to 40)
7	Generativity vs. stagnation	Care	Adult hood(40 to 65)
8	Ego integrity vs. despair	Wisdom	Maturity (65+)

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- **Attachment Theory**
- • **Definition of Attachment**
- • An enduring emotional tie to a special person, characterized by a tendency to seek and maintain closeness, especially during times of stress.

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- • Infant’s innate behaviors are evolved responses which promote survival
- • Affects internal working model
- **Bowlby’s Four Stages of Attachment**
- **Pre-attachment phase**
- • Birth - 6 weeks
- • Baby’s innate signals attract caregiver
- • Caregivers remain close by when the baby responds positively
- **Attachment in the Making**
- • 6 wks to 6-8 months
- • Develops a sense of trust that caregiver will respond when signaled
- • Infants respond more positively to familiar caregiver
- • Babies don't protest when separated from parent
- **Clear-cut Attachment**
- • 6-8 months to 18-24 months
- • Babies display separation anxiety
- • Babies protest when parent leaves
- **Formation of Reciprocal Relationship**
- • 18 month - 2yrs
- • Toddlers increase their understanding of symbols and language improves
- • Toddlers understand that parents will return
- **Factors which Affect Attachment**
- • Opportunity for attachment
- • Quality of caregiving
- - respond promptly and consistently
- - interactional synchrony – the sensitively tuned “emotional dance”
- • Infant characteristics
- -infant's temperament, special needs, prematurity, or illnesses
- • Family circumstances
- - Stress can undermine attachment
- • Parents’ internal working models
- - Parents’ own attachment experiences
- - Parents’ ability to accept their past
- **Measuring the Quality of Attachment**

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- • Mary Ainsworth (December 1, 1913 - March 21, 1999) was a developmental psychologist best known for her Strange Situation assessment and contributions to the area of attachment theory.
- • Ainsworth elaborated on Bowlby's research on attachment and developed an approach to observing a child's attachment to a caregiver.
- • This commonly known as Strange Situation
- **Mary Ainsworth strange situation**
- • Ainsworth researched using “strange situation”
- • It was a lab experiment with 8 different episodes of separation and reunion

Ainsworth styles of Attachment

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Ainsworth styles of Attachment

- • Based on her research, she identified four major styles of attachment that children have to their parents or caregivers.
- • These are:
- • ***Secure Attachment***
- • ***Anxious-Resistant Insecure Attachment***
- • ***Anxious-Avoidant Insecure Attachment***
- • ***Disorganized/Disoriented Attachment***:
- ***Secure Attachment***:
- • This happens when a child is very attached to the mother.
- • Usually this child will explore and engage with others when the mother is in the room, but, when the mother leaves, this child will show negative emotions.
- • When left alone with the stranger, the child will avoid him.
- ***Anxious-Resistant Insecure Attachment***:
- • This child becomes irritated when the stranger appears in the room, while mother is still near by and when mother leaves, the child cannot explore and continue play.

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- • He is very distressed. However, when mother comes back, child acts resentful and upset to the mother, he tries to move away from her.
- ***Anxious-Avoidant Insecure Attachment***:
- • This child seems careless.
- • He doesn't show much emotions whether mother is in the room or not.
- • He doesn't want to be played with or held. He acts the same with the stranger as well.
- ***Disorganized/Disoriented Attachment***:
- • This child might be distressed when the mother leaves the room and be relieved when she comes back. However, the child may not want to be held or may show anger once the mother approaches.
- • He might hit or rock.
- • Further research revealed that more than half of the mothers with a child who fell into this category had suffered a trauma immediately before the birth of the child and had developed depression because of that trauma.
- **Kohlberg's Stages of Moral Development**
- • Kohlberg's theory includes three levels of moral reasoning. These are:
- • **Level 1**: Pre-Conventional morality,
- • **Level 2**: Conventional Morality,
- • **Level 3**: Post-Conventional morality.
- • NB:Each of these levels are further divided into two stages, making for a total of six stages
- **Level 1 – Pre-Conventional**
- • It's the first level, in children between the ages of 4 and 10 years old.
- • This level consists of Stage 1 and Stage 2.
- • ***Stage1: "orientation toward punishment and obedience."*** Children in this stage tend to obey rules only to avoid punishment.
- • ***Stage 2: "instrumental purpose and exchange."*** children's actions are based on consideration for what others can do for them. They follow rules simply out of self-interest .

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- **Level 2 - Conventional**
- • The second level, achieved between ages 10 and 13, though some individuals never move beyond this level.
- • This level includes Stage 3 and Stage 4.
- • ***Stage 3*** : ***"maintaining mutual relations, approval of others, and the golden rule."***
- • children evaluate acts according to the motives behind them and can take circumstances into account. They want to help others, can judge the intentions of others, and can develop their own ideas regarding morality.
- • ***Stage 4 :"social concern and conscience."***
- • At this stage, individuals are concerned with respecting authority, maintaining social order, and doing their duty within society.
- • In this stage, an act is considered wrong if it harms others or violates a rule or law.
- **Level 3 – Post-Conventional**
- • The final level, reached in early adolescence or young adulthood, though some never reach this level.
- • comprised of Stage 5 and Stage 6.
- • **Stage 5** :**"morality of contract, of individual rights, and of democratically accepted law."I**ndividuals value the will of the majority and the well-being of society.
- • Though individuals at this stage can recognize that there are times when human need and the law are conflicted, they believe that it is better if people simply follow the law.
- • ***Stage 6:"morality of universal ethical principles."*** Individuals do what they think is right, even if it is in conflict with the law. They act according to their internalized standards of morality .
- • Because so few people attain Level 3, Kohlberg questioned the validity of this level, though he later proposed an additional seventh stage, which he described as the "cosmic" stage, in which individuals are able to consider the effect of their actions on the universe as a whole.

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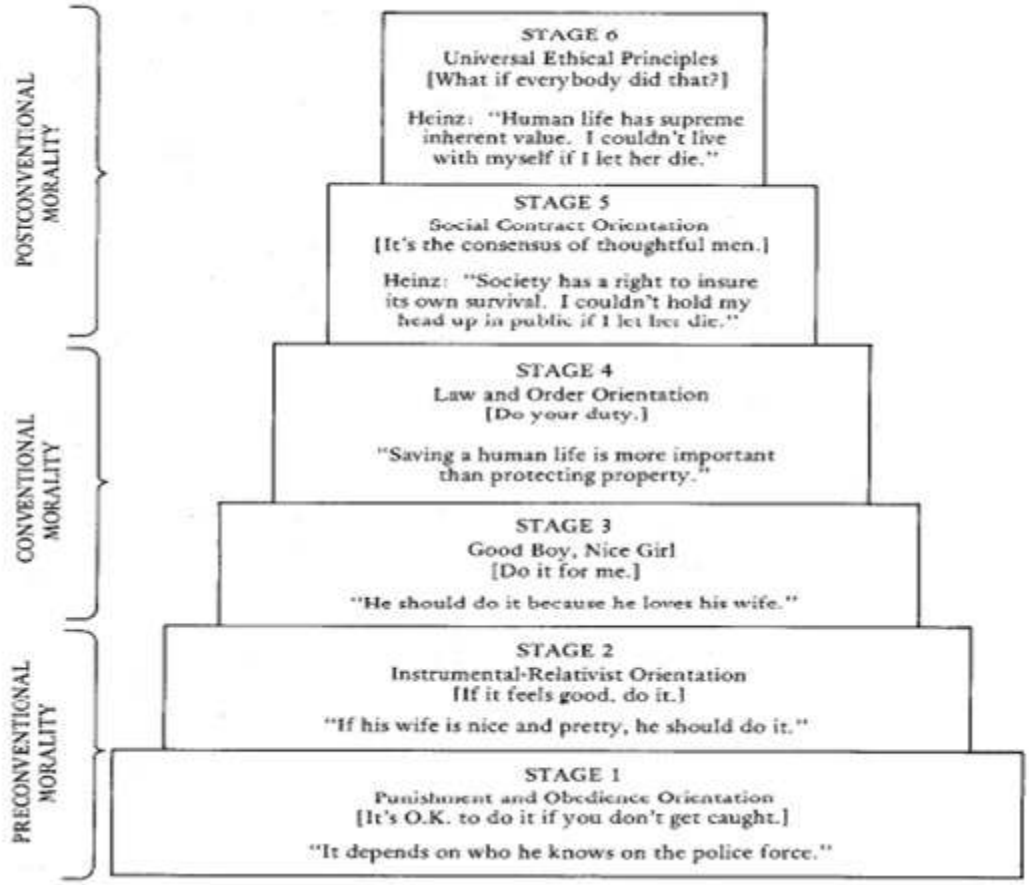

Gilligan's Theory Of moral development

- • Gilligan challenged Kohlberg's claim that all moral reasoning is "justice reasoning."
- • Gilligan argued that women's moral judgments include feelings of compassion and empathy for others, as well as concern for commitments that arise out of relationships.
- • Women engage in "care reasoning," not "justice reasoning," and thus consider their own and other's responsibilities to be grounded in social context and interpersonal commitments.
- • According to Gilligan, the male voice emphasizes independence ("separation") and responsibility for oneself.
- • Whereas the female voice emphasizes interdependence ("connection") and responsibility to others.
- • Males are encouraged to be active agents, females to be passive recipients.

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- • When faced with moral problems, males seek solutions that are just and fair; females seek solutions that are caring and compassionate.
- • For males, moral wrongness is linked to the violation of rights and justice; whereas for females, its linked to a failure to communicate and to respond.
- • For males, moral interactions take place primarily at the political and legal level, in the realm of abstract laws and social contracts; for females, moral interactions take place primarily at the level of personal relationships, in the family and the social network of the community in which they live.
- • Gilligan identified the following stages of moral development.
- • **Level 1**: Self-Oriented
- • **Level 2**: Other Oriented
- • **Level 3**: Universal Oriented
- ***Level 1: Self-Oriented***
- • Focus is on the needs of oneself.
- • Survival of oneself is of sole concern.
- • The transition to level 2 begins with the recognition of the conflict between one's own needs and the needs of others (i.e., what one owes to oneself vs. what one owes to others).
- ***Level 2: Other Oriented***
- • Focus is on the needs of others.
- • Self-adopts the traditional conception of feminine goodness, the maternal morality of self-sacrifice.
- • The good is equated with caring for others.
- • Consequently, one's own needs become devalued.
- • The transition to level 3 begins with the recognition that the self cannot be left out, but must also be an object of one's caring.
- ***Level 3: Universal Oriented***
- • Focus is on the universal obligation of caring.
- • Care is a self-chosen principle that condemns exploitation, violence, and neglect and demands active response to suffering.
- • Caring for oneself and others is seen as intertwined because the self and others are recognized as interdependent.
- Thus, all acts of caring are seen as beneficial to both self and others.

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Topic 1: Summary

We have described the following theories;

1. Erickson psychosocial theory of development.
2. Attachment theories by John Bowlby and Mary Ainsworth.
3. Moral development theory by Kohlberg.
4. Gilligan's stages of moral development

Topic 2: Life stages

At the end of this topic, the learner shall be able to;

•

1. Describe the four domains of human development
2. Explain the factors influencing human growth and development.
3. Identify the stages of human growth and development.
4. Discuss the life stages from conception to adolescence of human development.

Life Stages:

•Prenatal –before birth
•Infancy-birInfancy-birt•Early childhood- 1 to 6 years
•Middle and Late childhood- 6 to 12 years
•Adolescence- 12 to 20 years
•Early adulthood - 20 to 40 years
•Middle adulthood- 40 to 65 years
•Late adulthood- 65 years and up

Domains of human growth and development.

- In these life stages, an individuals experiences **four main type/ domains** of growth and development :
- **Physical**: refers to body growth.

It involves height and weight changes, muscle and nerve development, and
changes in body organs

- **Mental**:

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-refers to development of the mind.

- it includes learning how to solve problems, make judgments, and deal with situations.

- **Emotional:**
  - refers to feelings it includes dealing with love, hate, joy, fear, excitement, and other similar feelings.
- **Social:**
  -refers to interactions and relationships with others.

- *Each stage has its own characteristics and has specific developmental tasks that an individual must master*

Factors influencing growth & Dev't

- **Heredity**: Heredity and genes certainly play an important role in the transmission of physical and social characteristics from parents to off-springs. Different characteristics of growth and development like intelligence, aptitudes, body structure, height, weight, color of hair and eyes are highly influenced by heredity.
- **Sex:** Sex is a very important factor which influences human growth and development. There is lot of difference in growth and development between girls and boys. Physical growth of girls in teens is faster than boys. Overall the body structure and growth of girls are different from boys.
- **Socioeconomic:** It has been seen that the children from different socioeconomic levels vary in average body size at all ages. The upper level families being always more advanced. The most important reasons behind this are better nutrition, better facilities, regular meals, sleep, and exercise. Family size also influences growth rate as in big families with limited income sometimes have children that do not get the proper nutrition and hence the growth
- **Nutritional**: Growth is directly related with nutrition. The human body requires an adequate supply of calories for its normal growth and this need of requirements vary with the phase of development. As per studies, malnutrition is referred as a

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large-scale problem in many developing countries. They are more likely to be underweight, much shorter than average, and of low height for age, known as **stunting.**

- If the children are malnourished, this slows their growth process. There are nine different amino acids which are necessary for growth and absence of any one will give rise to stunted growth. Other factors like zinc, Iodine, calcium, phosphorus and vitamins are also essential for proper growth and deficiency of anyone can affect the normal growth and development of the body.
- **Pollution**: According to studies, air pollution not only affects the respiratory organs but also have harmful effects on human growth.

Indoor pollution or the pollution from housing conditions can result in ill health which can negatively impact human growth and development. For example, lead exposure from deteriorated lead-based paint in older housing can be very harmful. Lead is very harmful for children as it simply gets immersed into the growing bodies of children and obstructs with the normal development of brain and other organs and systems.

- **Race:** Racial factors also influence height, weight, color, features, and body constitution of a human being. The body growth and development differences show a relationship with varied cultural groups. For example a child of black race will be black, their height, their hair and eye color, facial structure are all governed by the same race.

Topic 2: Prenatal Development

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- **Prenatal Development**

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- • Prenatal development is the process that occurs during the 40 weeks prior to the birth of a child, and is heavily influenced by genetics.
- • organized into trimesters: the first trimester ends with the end of the embryonic stage, the second trimester ends at week 26, and the third trimester ends at birth.
- • There are three stages
- • *germinal, embryonic,* and *fetal*
- • The germinal stage conception until 2 weeks (implantation), during which the zygote begins to rapidly divide.
- • The embryonic stage from implantation (2 weeks) until the 8th week of pregnancy.
- • During this stage, rapid growth occurs and organ systems develop.
- The fetal stage lasts from week 9 until the child's birth (usually between 38 and 40 weeks). Throughout this stage the brain continues to grow and develop at a rapid pace

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Factors that influence Prenatal Development

- • During prenatal development, there are factors that significantly affect the development of the child.
- • almost everything the mother ingests, including food, liquid, and even medication, travels through the placenta to the fetus; anything the mother is exposed to in the environment affects the fetus.
- • A teratogen is any substance or agent in the environment that can have a detrimental effect on a developing fetus. Include drugs, alcohol, tobacco, and other environmental agents.
- Infancy: Birth to 1 year
- Physical Development
- • The most dramatic and rapid changes in growth and development occurring during this time.
- • Newborn usually weighs: 2 to 4kgs
- • Measures: 18-22 inches
- • End of year 1:
- -Weight has usually tripled, to 6 to 12kgs
- -Height has increased to approx. 29 to 30 inches.
- • Muscular and nervous system developments are dramatic.
- • Motor or startle reflex to a loud noise or sudden movement.
- • Rooting reflex, in which a slight touch on the cheek causes the mouth to open and the head to turn.
- • Sucking reflex, caused by a slight touch to the lips.
- • Grasp reflex, in which infants grasp an object placed in the hand.
- • Muscle coordination develops in stages:
- • At first, infants are able to lift the head slightly
- 2 months:
- • they can usually roll from side to back
- 4 to 5 months:
- • Turn the body completely around
- • Accept objects handed to them
- • Grasp stationary objects
- • With support, hold the head up while sitting
- 6 to 7 months:
- • Can sit unsupported for several minutes

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- • Grasp moving objects
- • Crawl on the stomach
- By 12 months:
- • Frequently can walk without assistance
- • Grasp objects with thumb and fingers
- • Throw small objects
- • Usually will have 10 to 12 teeth by the end of the first year of life.
- • At birth vision is poor and may be limited to black and white, and eye movements are not coordinated. By 1 year, close vision is good, in color, and can readily focus on small objects.
- • Sensory Abilities:
- -Smell
- -Taste
- -Sensitivity to hot and cold
- -Hearing
- *-Good at birth, become more refined and exact*

- **Infancy**
- **Mental Development:**
- • Newborns respond to discomforts such as pain, cold, or hunger by crying.
- • When needs are met, they become more aware of their surroundings and people.
- • Responds to stimuli in the environment and learning activities grow.
- • By 6 months, understand some words and can make basic sounds
- • By 12 months understand many words and use many single words in their vocabularies.
- **Emotional Development**
- • Observed very early in life
- • Newborns show excitement
- • 4 to 6 months:
- -Distress

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- -Anger
- -Disgust
- -Fear
- *These can often be seen.*
- By 12 months of age:
- • Elation and affection for adults is evident
- **<u>Social Development</u>**
- • Goes from self-centeredness to recognition of others in the environment.
- By 4 months of age:
- • Recognize their caregivers
- • Smile readily
- • Stare intently at others
- By 6 months of age:
- • Infants watch the activities of others
- • Show signs of possessiveness
- • May become shy or withdraw when in the presence of strangers
- By 12 months:
- • May still be shy with strangers
- • Socialize freely with familiar people
- • Mimic and imitate gestures
- • Facial expressions
- • Vocal sounds
- **Things to remember:**
- • Infants are dependent on others for all needs
- • Food, cleanliness, and rest are essential for physical growth.
- • Love and security are essential for emotional and social growth.
- • Stimulation is essential for mental growth.
- **Red Flags in infant development**
- • Unable to sit alone by age 9 months
- • Unable to transfer objects from hand to hand by age 1 year
- • Abnormal pincer grip or grasp by age 15 months
- • Unable to walk alone by 18 months
- • Failure to speak recognizable words by 2 years.
- **Early Childhood: 1 to 6 years of age**
- **<u>Physical Development:</u>**

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- • Physical growth is slower
- • By age 6, the average weight is 21kgs and the average height is 46 inches
- • Skeletal and muscle development helps the child assume a more adult appearance.
- • Legs and lower body tend to grow more rapidly
- • Muscle Coordination:
- -Run
- -Climb
- -Move more freely
- • Muscles of the fingers develop:
- -Write
- -Draw
- -Use a fork and knife
- • By 2 or 3yrs, most teeth have erupted
- • The digestive system is mature enough for most adult foods.
- • Between 2 and 4 years of age, most children learn bladder and bowel control
- **Mental Development:**
- • Advances rapidly during this stage
- • Verbal growth progresses from the use of several words to a vocabulary of 1,500 to 2,500 words by age 6.
- • 2-year olds have short attention span, but are interested in many different activities.
- • Remember details and understand concepts
- • 4-year olds ask frequent questions and usually recognize letters and some words.
- • Decisions based on logic rather than on trial and error.
- • By age 6:
- • Very verbal
- • Want to learn to read and write
- • Memory developed to the point where the child can make decisions based on both past and present experiences.
- **Emotional Development:**
- • Advances rapidly
- • Ages 1 to 2, children begin to develop self-awareness and recognize the effect they have on other people and things.
- • 1 to 2 years of age:

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- • Limits are usually established for safety.
- • Will usually accept or defy limits.
- • Feel impatience and frustration as they try to do things beyond their abilities.
- • Anger, often in the form of “temper tantrums”, occurs when they cannot perform as desired.
- • 2 to 4 years of age:
- • Like routines
- • Become stubborn, angry, or frustrated when change occurs.
- Age 4 to 6
- • Gain more control over their emotions.
- • Understand the concept of right and wrong.
- • Because they have achieved some independence, not as frustrated as much by their lack of ability.
- By age 6:
- • Show less anxiety when faced with new experiences, because they have learned they can deal with new situations.
- **Social Development:**
- • Expands from a self-centered 1-year-old to a very sociable 6-year-old.
- • Are strongly attach to their parents and they fear any separation.
- • Enjoy company of others, but are still very possessive.
- • Put “self” aside and begin to take more of an interest in others.
- • Trust other people and make more of an effort to please others by becoming more agreeable and social.
- • Friend of their own age are usually more important to 6-year-olds
-
- **Early Childhood**
- **Needs:**
- -Food
- -Rest
- -Shelter
- -Protection
- -Love
- -Security

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- **Also need:**
- -Routine
- -Order
- -Consistency in their daily lives
- • They must be taught to be responsible and must learn how to conform to rules.
- • This can be accomplished by making reasonable demands based on the child's ability to comply.
- **Red flags:Early childhood /preschool**
- • Inability to perform simple self-care tasks (*e.g. hand washing, simple dressing, daytime toileting)*
- • Lack of socialization
- • Unable to play with other children
- • Unable to follow directions during exam
- **Middle and Late Childhood**
- Ages 6 to 12 (preadolescence)
- **<u>Physical Development:</u>**
- • Slow but steady
- • Weight gain averages 2.5 to 3.2kgs per year
- • Height usually increases approximately 2 to 3 inches per year.
- • Muscle coordination is well developed
- • Engage in physical activity that requires complex motor-sensory coordination.
- • Most of the primary teeth are lost and permanent teeth erupt
- • Eyes are well developed and visual acuity is at its best
- • During ages 10 to 12 sexual maturation may begin in some children
- **<u>Mental Development</u>**
- • Increases rapidly because the child's life centers around school.
- • Speech skills develop more completely.
- • Reading and writing skills are learned.
- • Use information to solve problems
- • Memory becomes more complex
- • Begin to understand concepts such as:
- -loyalty
- -honesty

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- -values
- -Morals
- • Use more active thinking
- • More adept to making judgments.
- **<u>Emotional Development:</u>**
- • Continues to help the child achieve a greater independence and a more distinct personality
- Age 6
- • Frightened and uncertain as they begin school
- • Reassuring parents and success in school help the child gain self-confidence
- • Fears are replaced by the ability to cope
- • Emotions are slowly brought under control
- 8 to 10 years of age
- • More group orientated
- • Form groups with members of their own sex
- • Ready to accept the opinions of others and learn to conform to rules and standards of behavior by the group
- Age 10 to 12
- • Sexual maturation and changes in body function can lead to periods of depression followed by periods of joy.
- • Changes can cause children to be restless, anxious, and difficult to understand
- • Toward the end of this period, tend to make friend more easily and awareness towards the opposite sex.
- • As they spend time with others their own age, their dependency on their parents lessens
- **Needs:**
- • Same basic needs of infancy and early childhood
- • Now needs:
- -reassurance
- -parental approval
- -peer acceptance
- **Red flags: Middle and Late Childhood /school age**
- • School failure
- • Lack of friends
- • Social isolation
- • Aggressive behavior: *fights, fire setting, animal abuse*

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-
- **Adolescence**
- • Ages 12 to 20
- **Physical development**
- • Traumatic life stage
- • Sudden “growth spurt”
- • Weight gain up to 10kg
- • Height increase of several inches can occur in a period of months
- • Growth spurt for girls ages 11 to 13
- • Growth spurt for boys ages 13 to 15
- • Muscle coordination does not advance as quickly.
- • Awkwardness or clumsiness
- Puberty:
- • Development of sexual organs and the secondary sexual characteristics.
- **Mental Development:**
- • Primarily involves an increase in knowledge and a sharpening of skills
- • Learn to make decisions and to accept responsibility for their actions.
- • Time of conflict:
- • Told to “grow up” while being reminded that they are “still children”.
- **Emotional Development:**
- • Stormy and in conflict.
- • Trying to establish identity and independence, but are often uncertain and feel inadequate and insecure.
- Worry about:
- -Appearance
- -Their abilities
- -Relationship with others
- • Frequently respond more and more to peer group influences.
- • This can lead to changes in attitude and behavior and conflict with values previously established.
- • Toward the end of adolescence, self-identity has been established.

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- • More comfortable with who they are and turn attention toward what they may become.
- • Gain more control of their feelings and become more mature emotionally
- **Social Development:**
- • Spending less time with family and more time with peers.
- • Seek security in groups of people their own age who have similar problems and conflicts.
- • Toward the end of this life stage, adolescents develop a more mature attitude and begin to develop patterns of behavior that they associate with adult behavior.
- **Needs:**
- -Reassurance
- -Support
- -Understanding
- • Many problems can develop during this life stage and can be traced to the conflict and feelings of inadequacy and insecurity that adolescents experience.
- **Eating Disorders:**
- • Often develop from an excessive concern with appearance.
- ***Anorexia Nervosa:***
- • Commonly called “Anorexia”
- • Psychological disorder in which a person drastically reduces food intake or refused to eat at all.
- • Leads to metabolic disturbances, excessive weight loss, weakness, and if not treated, death.
- ***Bulimia:***
- • Psychological disorder
- • Person alternately binges and fasts, or refuses to eat at all.
- • When a person induces vomiting or uses laxatives to get rid of food that has been eaten, this condition is called “*bulimarexia*”.
- **Substance Abuse:**
- • The use of a substance such as alcohol or drugs and development of a physical and/or mental dependence on these substances.
- • Can occur in any life stage, but frequently begins in adolescence.
- **Reasons for using chemicals:**

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- -Anxiety
- -Stress relief
- -Peer pressure
- -Escape from emotional or psychological problems
- -Experimentation with feelings the chemical produces
- -Desire for “instant gratification”
- -Hereditary traits
- -Cultural influences
- • Substance abuse can lead to physical and mental disorders and disease.
- • Treatment is directed toward total rehabilitation that allows the substance abuser to return to a productive and meaningful life.
- **Suicide**
- • Found in many life stages, is one of the leading causes of death in adolescents.
- • Always a permanent solution to a **temporary** problem.
- **Reasons?**
- • Depression
- • Grief over a loss or love affair
- • Failure in school
- • Inability to meet expectations
- • Influence of suicidal friends
- • Lack of self-esteem
- **The risk of suicide increases:**
- • Family history of suicide
- • Major loss or disappointment
- • Previous suicide attempts
- • Recent suicide of friends, family, or role models
- • Impulsive nature of adolescents
- Most individuals who are thinking of suicide give warning signs such as verbal statements:
- • “I’d rather be dead”
- • “You’d be better off without me”
- Other warning signs include:
- • Sudden change in appetite and sleep habits
- • Withdrawal, depression, and moodiness
- • Neglect of personal hygiene

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- • Alcohol or drug abuse
- • Losing interest in hobbies and other aspects of life
- • Giving away possessions
- • Saying goodbye to family and friends
- *These individuals are calling out for attention and help and usually respond to efforts of assistance.*
- • ignored.
- • Support, understanding, and psychological or psychiatric counseling are used to prevent suicide.

Topic 2: Summary

-

In this topic, we have been able to;

-

1. Describe the four domains of human development
2. Explain the factors influencing human growth and development.
3. Identify the stages of human growth and development.
4. Discuss the life stages from conception to adolescence of human development.

Topic three: Adulthood

By the end of this topic, the learner should be able to;

1. Describe the characteristics of the various stages of adulthood
2. Describe old age and aging
3. Describe the process of death and dying

Introduction

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🎬 **Aging factors**

🎬 • Each person experiences age-related changes in adulthood.

🎬 • based on many factors: biological factors such as molecular and cellular changes are called ***primary aging***, while aging that occurs due to controllable factors, such as lack of physical exercise and poor diet, is called ***secondary aging*.**

🎬 **Early Adulthood (Ages 20–40)**

🎬 • The aging process, although not obvious, begins.

🎬 • Around the age of 30, many changes begin to occur in different parts of the body.

🎬 • For example, the lens of the eye starts to stiffen and thicken, resulting in changes in vision (usually affecting the ability to focus on close objects).

🎬 • Sensitivity to sound decreases; this happens twice as quickly for men as for women.

🎬 • Hair can start to thin and become gray around the age of 35, earlier for some individuals and later for others.

🎬 • The skin becomes drier and wrinkles start to appear by the end of early adulthood.

🎬 • The immune system becomes less proficient at fighting off illness, and

🎬 reproductive capacity starts to decline.

🎬 **Cognitive Development**

🎬 • Cognitive changes over a person's lifespan, peak at around age 35 and slowly decline in later adulthood.

🎬 • Because we spend so many years in adulthood (more than any other stage), cognitive changes are numerous during this period.

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- Research suggests that adult cognitive development is a complex, ever-changing process that may be even more active than cognitive development in infancy and early childhood (Fischer, Yan, & Stewart, 2003).
- Our cognitive abilities remain relatively steady throughout early and middle adulthood.
- Research has found that adults who engage in mentally and physically stimulating activities experience less cognitive decline in later adult years .
- Have a reduced incidence of mild cognitive impairment and dementia.

Beyond Piaget's Theory

- According to Jean Piaget's theory of cognitive development, the establishment of **formal operational** thinking occurs during early adolescence and continues through adulthood.
- This kind of thinking is characterized by the ability to think in abstract ways, engage in deductive reasoning and create hypothetical ideas to explain various concepts.
- Since Piaget's theory, other developmental psychologists have suggested a fifth stage of cognitive development, known as ***postformal operational thinking***.
- In postformal thinking, decisions are made based on situations and circumstances, and logic is integrated with emotion as adults develop principles that depend on contexts
- This kind of thinking includes the ability to think in dialectics, and differentiates between the ways in which adults and adolescents are able to cognitively handle emotionally charged situations
- Its a time of relativistic thinking, in which young people begin to become aware of more than simplistic views of right vs. wrong.

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🎬 • They begin to look at ideas and concepts from multiple angles and understand that a question can have more than one right (or wrong) answer.

🎬 • The need for specialization results in pragmatic thinking—using logic to solve real-world problems while accepting contradiction, imperfection, and other issues.

🎬 • Finally, young adults develop a sort of expertise in either education or career, which further enhances problem-solving skills and the capacity for creativity.

🎬 • Physical maturation is complete, however height and weight may increase slightly.

🎬 • Physical abilities are at their peak, i.e muscle strength, reaction time, sensory abilities, and cardiac functioning.

🎬 • Most professional athletes are at the top of their game

🎬 Many women have children in the early-adulthood years

🎬

- **Middle adulthood(Ages 40–65)**
- **Physical development**
- • the aging process becomes more apparent.
- • Around the age of 60, the eyes lose their ability to adjust to objects at varying distances, known as **presbyopia**.
- • Most people between the ages of 40 and 60 will need some form of corrective lenses for vision deficits.
- • Middle-aged adults are also at higher risk than younger adults for certain eye problem, such as glaucoma.
- • Hearing also further declines.
- • Women experience a gradual decline in fertility as they approach the onset of menopause—the end of the menstrual cycle—around 50 years old.
- • This process involves hormonal changes and may last anywhere from six months to five years.
- • Due to the shifting hormone levels, women going through menopause often experience a range of other symptoms, such

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as anxiety , poor memory, inability to concentrate, depressive mood, irritability, mood swings, and less interest in sexual activity. Among other physiological symptoms.

- • Two forms of intelligence—crystallized and fluid—are the main focus of middle adulthood.
- • Our ***crystallized intelligence*** is dependent upon accumulated knowledge and experience—it is the information, skills, and strategies we have gathered throughout our lifetime.
- • This kind of intelligence tends to hold steady as we age—in fact, it may even improve. For example, adults show relatively stable to increasing scores on intelligence tests until their mid-30s to mid-50s.
- • ***Fluid intelligence,*** on the other hand, is more dependent on basic information-processing skills and starts to decline even prior to middle adulthood.
- • Cognitive processing speed slows down during this stage of life, as does the ability to solve problems and divide attention.
- • However, *practical* problem-solving skills tend to increase. These skills are necessary to solve real-world problems and figure out how to best achieve a desired goal.
- • **Social and Emotional Aspects of Adulthood**
- • There are many theories about the social and emotional aspects of aging. Some aspects of healthy aging include activities, social connectedness, and the role of a person's culture.
- • According to many theorists, including George Vaillant (2002), who studied and analyzed over 50 years of data, we need to have and continue to find meaning throughout our lives.
- • For those in early and middle adulthood, meaning is often found through work (Sterns & Huyck, 2001) and family life .
- • These areas relate to the tasks that Erik Erikson referred to as *generativity vs. stagnation* and *intimacy vs. isolation*
- **Relationships in Adulthood**
- • Positive relationships with significant others in our adult years have been found to contribute to a state of well-being.
- • Most adults identify themselves through their relationships with family—particularly with spouses, children, and parents.

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- • While raising children can be stressful, especially when they are young, research suggests that parents reap the rewards down the road, as adult children tend to have a positive effect on parental well-being .
- • Having stable intimate relationships has also been found to contribute to well-being throughout adulthood.
- • A lack of positive and meaningful relationships during adulthood can result in what Erikson termed the crisis of *intimacy vs. isolation* in his theory of psychosocial development.
- • In young adulthood (i.e., 20s and early 30s), people tend to be concerned with forming meaningful relationships.
- • young and middle-aged adults are subject to loneliness if they are unable to form meaningful relationships with family, friends, or community.
- • Both early and middle adulthood come with particular challenges; these challenges are at times referred to as "quarter-life crises" and ‘**’midlife crises**," respectively.
-
- **Crises of Adulthood**
- • A **quarter-life crisis** typically occurs between the ages of 25 and 30.
- • often revolves around the challenges that arise from young adults newly living life on their own and feeling overwhelmed with new responsibilities;
- • Can also happen after the birth of a child or if a person graduates from college and cannot find a job in their chosen field.
- • In this stage of life, young people may worry about their future, wonder if they've made poor choices, or wonder what life might hold for them now.
- • The main triggers for **a mid-life crisis** include problems with work, trouble in a marriage, children growing up and leaving the home, or the aging or death of a person's parents.
- • This is likely to occur during Erikson's stage of ***generatively vs. stagnation***, a time when people think about the contribution they are making to the world.

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- • Generatively involves finding one's life's work and contributing to the development of others through activities such as volunteering, mentoring, and raising children;
- • those who do not master this task may experience a feeling of stagnation.
- Individuals having a mid-life crisis may experience some of the following:
- -A search for an undefined dream or goal;
- -A deep sense of regret for goals not accomplished;
- -A fear of humiliation among more successful colleagues;
- -A desire to achieve a feeling of youthfulness;
- -A need to spend more time alone or with certain peers.
- • Some who experience a quarter- or mid-life crisis struggle with how to cope and may engage in harmful behaviors, such as abuse of alcohol or drugs or excessive spending of money.
- • Others may experiment with different aspects of their personality, explore new hobbies, or otherwise seek out change in their lives
- • Many adults find meaning in and define themselves by what they do—their careers.
- • Earnings peak for many during adulthood.
- • However research has found that job satisfaction is more closely tied to work that involves contact with other people, is interesting, provides opportunities for advancement, and allows some independence than it is to salary.
- • Erikson's stage of *generativity vs. stagnation* revolves around a person's sense of their contribution to the world.
- • Generativity is about making life productive and creative so that it matters to others, especially those in the next generation.
- • According to Erikson, a person who is self-centered and unable or unwilling to help society move forward develops a feeling of stagnation—a dissatisfaction with the relative lack of productivity.
- • The central tasks during middle adulthood can include expressing love through more than sexual contacts, maintaining healthy life patterns, helping growing and grown children to be responsible adults, relinquishing a central role in the lives of grown children, creating a comfortable home, being proud of

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one's accomplishments, taking care of aging parents, adjusting to the physical changes of middle age, and using leisure time creatively.

- **Interpersonal Relationships in Adulthood**
- • Positive relationships with significant others in our adult years have been found to contribute to a state of well-being.
- • An interpersonal relationship is a strong, deep, or close association or acquaintanceship between two or more people that may range from brief to enduring in duration.
- • Like people, relationships change and grow; they may either improve or dissipate over time. The association between two people can be based on various factors—love, solidarity, business, or any other context that requires two (or more) people to interact.
- • Interpersonal relationships are dynamic systems that change continuously during their existence. Like living organisms, relationships have a beginning, a lifespan, and an end. They tend to grow and improve gradually as people get to know each other and become closer emotionally, or they gradually deteriorate as people drift apart.
- **Levinger's Model of Relationships**
- • One of the most influential models of relationship development was proposed by psychologist George Levinger.
- • The model was formulated to describe heterosexual, adult romantic relationships, but it has been applied to other kinds of interpersonal relationships since then.
- According to the model, the natural development of a relationship follows five stages:
- • ***Acquaintance and Acquaintanceship***: Becoming acquainted depends on previous relationships, physical proximity, first impressions, and a variety of other factors. If two people begin to like each other, continued interactions may lead to the next stage, but acquaintance can continue indefinitely.
- • Another example is association.
- • ***Buildup***: During this stage, people begin to trust and care about each other.

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- • The need for intimacy, compatibility, and such filtering agents as common background and goals will influence whether or not interaction continues.
- • ***Continuation*:** This stage follows a mutual commitment to strong and close long-term friendships, romantic relationship, or even marriage.
- • It is generally a long, relatively stable period. Nevertheless, continued growth and development will occur during this time.
- • Mutual trust is important for sustaining the relationship.
- • ***Deterioration***: Not all relationships deteriorate, but those that do tend to show signs of trouble.
- • Boredom, resentment, and dissatisfaction may occur.
- • Individuals may communicate less and avoid self-disclosure.
- • Loss of trust and betrayals may take place as the downward spiral continues, eventually ending the relationship.
- • Alternately, the participants may find some way to resolve the problems and reestablish trust.
- • ***Termination*:** The final stage marks the end of the relationship, either by breakup, death, or spatial separation and severing all existing ties of either friendship or romantic love.
- **Knapp's theory of Relational Development**
- • Another theory, developed by M. L. Knapp, known as the **model of relational development**.
- • It consists of two main stages, each with several parts.
- • The first stage known as the ***coming together* phase,**
- • the second stage known as the ***coming apart* phase**
- **Knapp's theory of Relational Development Coming Together**
- • Coming together consists of five phases
- • *initiating,*
- • *experimentation,*
- • *intensifying,*
- • *integration*, and
- • *bonding.*
- • During *initiating*, first impressions are made; physical factors play a large role in this phase. People often want to portray themselves as easy to talk to, friendly, and open to discussion.

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This phase tends to be superficial as people are trying to make a good first impression.

- • During *experimentation*, the two people attempt to find some common ground between each other's lives, such as common interests and hobbies. People start to open up more and ask more personal questions as they get to know one another.
- • During the *intensifying* phase, people open themselves up fully in the hope of being accepted by the potential mate. During this phase, people may reveal secrets about themselves or others in order to test the trust level of potential partners
- • The *integration* phase involves people merging their lives together and solidifying a relationship status.
- • The *bonding* phase, people recognize a commitment to one another (traditionally through marriage, though many alternative forms of commitment exist) and the relationship lasts until death, breakup, or divorce.
- • **Knapp's theory of Relational Development Coming Apart**
- • Coming apart consists of five stages:—
- • *differentiating,*
- • *circumscribing,*
- • *stagnation,*
- • *avoidance*, and
- • *termination.*
- • *Differentiation* involves focusing more on differences rather than similarities. This can lead to an increasing emotional distance between the parties involved.
- • During *circumscribing*, the primary focus of the relationship shifts from differences to setting limits and boundaries on communication between the two people. This further pushes two people apart.
- • *Stagnation* is when two people have reached a "stand-off" phase—nothing changes and neither party is willing to change.
- • *Avoidance* occurs when people engage in limited communication and take steps to distance themselves from one another.
- • The *termination*, the relationship is ended.
- **Old age/Late Adulthood**

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- • **Physical Development**
- • The aging process often results in a loss of memory, deteriorated intellectual function, decreased mobility, and higher rates of disease.
- **Physical Changes**
- • The stage of life from the 60s onward;
- • Constitutes the last stage of physical change.
- • Average life expectancy varies greatly based on factors such as socioeconomic status, region, and access to medical care.
- • Women tend to live longer than men by an average of five years.
- • The skin continues to lose elasticity,
- • Reaction time slows further,
- • Muscle strength diminishes.
- • Hearing and vision decline significantly;
- • Cataracts, or cloudy areas of the eyes that result in vision loss, are frequent.
- • Other senses, such as taste, touch, and smell, get to be less sensitive .
- • The immune system is weakened,
- • More susceptible to illness, cancer, diabetes, and other ailments.
- • Cardiovascular and respiratory problems become more common.
- • A decrease in physical mobility and a loss of balance, which can result in falls and injuries.
- **Changes in the Brain**
- • The aging process results in changes and lower functioning in the brain.
- • This leads to problems like memory loss and decreased intellectual function. Age is a major risk factor for most common neurodegenerative diseases, including mild cognitive impairment, Alzheimer's disease, cerebrovascular disease, Parkinson's disease, and Lou Gehrig's disease.
- • Molecular changes are due in part to a reduction in the size of the brain, as well as loss of brain plasticity.

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- • *Brain plasticity* is the brain's ability to change structure and function.
- • The brain's main function is to decide what information is worth keeping and what is not; if there is an action or a thought that a person is not using, the brain will eliminate space for it.
- • Brain size and composition change along with brain function.
- • Computed tomography (CT) studies have found that the cerebral ventricles expand as a function of age in a process known as *ventriculomegaly.*
- • More recent MRI studies have reported age-related regional decreases in cerebral volume.
- • The brain begins to lose neurons in later adult years;
- • The loss of neurons within the cerebral cortex occurs at different rates, with some areas losing neurons more quickly than others.
- • The frontal lobe (which is responsible for the integration of information, judgment, and reflective thought) and corpus callosum tend to lose neurons faster than other areas, such as the temporal and occipital lobes.
- • The cerebellum , which is responsible for balance and coordination, eventually loses about 25 percent of its neurons as well.
- **Cognitive development**
- • A general decline in memory is very common, due to the decrease in speed of encoding, storage, and retrieval of information.
- • This can cause problems with short-term memory retention and with the ability to learn new information.
- • In most cases, this absent-mindedness should be considered a natural part of growing older rather than a psychological or neurological disorder.
- **Changes in Memory**
- • Memory also degenerates with age, and older adults tend to have a harder time remembering and attending to information. In general, an older person's procedural memory stays the same, while working memory declines.
- • **Changes in Memory**

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- • *Procedural memory* is memory for the performance of particular types of action; it guides the processes we perform and most frequently resides below the level of conscious awareness.
- • *Working memory* is the system that actively holds multiple pieces of transitory information in the mind where they can be manipulated. The reduced capacity of the working memory becomes evident when tasks are especially complex.
- • *Semantic memory* is the memory of understanding things, of the meaning of things and events, and other concept-based knowledge.
- • This type of memory underlies the conscious recollection of factual information and general knowledge about the world, and remains relatively stable throughout life.
- • Distinct from a normal decline in memory is dementia,
- • a broad category of brain diseases that cause a gradual long-term decrease in the ability to think and remember to the extent that a person's daily functioning is affected.
- • While the term **"dementia**" is still often used in lay situations, in the DSM-5 it has been renamed "neurocognitive disorder," with various degrees of severity.
- **Changes in Memory**
- • Neurocognitive disorders most commonly affect memory, visual-spatial ability, language, attention, and executive functions(e.g., judgment and problem-solving).
- • Most of these disorders are slow and progressive; by the time a person shows signs of the disease, the changes in their brain have already been happening for a long time.
- • About 10% of people with dementia have what is known as *mixed dementia*, which is usually a combination of Alzheimer's disease and another type of dementia.
- • *Alzheimer's disease* is the most common type of neurocognitive disorder, accounting for 50% to 70% of cases.
- • There is no cure for dementia, but for people who suffer from these disorders and for their caregivers, many measures can be taken to improve their lives.

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- • These can include education and support for the caregiver and daily exercise programs or cognitive or behavioral therapies for the person with the disorder.
- **Socio-emotional Development .**
- • **Increased Dependency**
- • Many elderly people need assistance in meeting daily needs as they age, and over time they may become dependent on caregivers such as family members, relatives, friends, health professionals, or employees of senior housing or nursing care.
- • Many older adults spend their later years in assisted living facilities or nursing homes, which can have social and emotional impacts on their well-being.(Developed Countries)
- • Older adults may struggle with feelings of guilt, shame, or depression because of their increased dependency, especially in societies where caring for the elderly is viewed as a burden.
- • If an elderly person has to move away from friends, community, their home, or other familiar aspects of their life in order to enter a nursing home, they may experience isolation, depression, or loneliness.
- • Increased dependency can also put older adults at risk of elder abuse.
- • This kind of abuse occurs when a caretaker intentionally deprives an older person of care or harms the person in their charge.
- • The elderly may be subject to many different types of abuse, including physical, emotional, or psychological.
- • Approximately one in ten older adults report being abused, and this number rises in the cases of dementia or physical limitations.
- • Despite the increasing physical challenges of old age, many new assistive devices made especially for the home have enabled more old people to care for themselves and accomplish activities of daily living (ADL).
- • Some examples of devices are a medical alert and safety system, shower seat (preventing the person from getting tired in the shower and falling), bed cane (offering support to those with unsteadiness getting in and out of bed), and ADL cuff (used with eating utensils for people with paralysis or hand weakness).

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- • Advances in this kind of technology offer increasing options for the elderly to continue functioning independently later into their lives.
-
- **Loneliness and Connection**
- • A central aspect of positive aging is believed to be social connectedness and social support. As we get older, **Socioemotional selectivity theory** suggests that our social support and friendships dwindle in number, but remain as close as, if not closer than, in our earlier years (Carstensen, 1992)..
- • Many older adults contend with feelings of loneliness as their loves ones, partners, or friends pass away or as their children or other family members move away and live their own lives.
- • Loneliness and isolation can have detrimental effects on health and psychological well-being.
- • Many adults counteract loneliness by having active social lives, living in retirement communities, or participating in positive hobbies. Staying active and involved in life counteracts loneliness and helps increase feelings of self-esteem and self-worth.
- **Erikson: Integrity vs. Despair**
- • As people enter the final stages of life, they have what Erik Erikson described as a crisis over *integrity versus despair.*
- • they review the events of their lives and try to come to terms with the mark (or lack thereof) that they have made on the world.
- • People who believe they have had a positive impact on the world through their contributions live the end of life with a sense of integrity.
- • Those who feel they have not measured up to certain standards—either their own or others'—develop a sense of despair
- **Confronting Death**
- • Kübler-Ross (1969), who worked with the founders of hospice care, described in her theory of grief the process of an individual accepting their own death. She proposed five stages of

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grief in what became known as the Kübler-Ross model: *denial, anger, bargaining, depression,* and *acceptance.*

- **Kübler-Ross model**
- • *Denial*: People believe there must be some mistake. They pretend death isn't happening, perhaps live life as if nothing is wrong, or even tell people things are fine. Underneath this facade, however, is a great deal of fear and other emotions.
- • *Anger*: After people start to realize death is imminent, they become angry. They believe life is unfair and usually blame others (such as a higher power or doctors) for the state of being they are experiencing.
- • *Bargaining*: Once anger subsides, fear sets in again. Now, however, people plead with life or a higher power to give them more time, to let them accomplish just one more goal, or for some other request.
- • *Depression*: The realization that death is near sets in, and people become extremely sad. They may isolate themselves, contemplate suicide, or otherwise refuse to live life. Motivation is gone and the will to live disappears.
- • *Acceptance*: People realize that all forms of life, including the self, come to an end, and they accept that life is ending. They make peace with others around them, and they make the most of the time they have remaining.
- **Kübler-Ross model**
- • While most individuals experience these stages, not all people go through every stage.
- • The stages are not necessarily linear, and may occur in different orders or reoccur throughout the grief process.
- • Some psychologists believe that the more a dying person fights death, the more likely they are to remain stuck in the denial phase, making it difficult for the dying person to face death with dignity.
- • However, other psychologists believe that not facing death until the very end is an adaptive coping mechanism for some people.
- • Whether due to illness or old age, not everyone facing death or the loss of a loved one experiences the negative emotions outlined in the Kübler-Ross model.

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- Research suggests that people with religious or spiritual beliefs are better able to cope with death because of their belief in an afterlife and because

Topic 3: Summary

-

In this topic,we have been able to;

1. Describe the characteristics of the various stages of adulthood
2. Describe old age and aging
3. Describe the process of death and dying

Unit Three Content..

Topic 1: Introduction to cognitive psychology

At the end of this topic, the learner should be able to

1. Define cognitive psycholog
2. Identify the major components
3. Discuss the terminologies in Thinking.
4. Discuss the thought processes in Thinking

- ***Cognitive Psychology-Definitions***
-
- Is the study of mental processes.
- • Is the study of thinking, learning, intelligence, memory and language in humans.
- • Is the scientific study of the human mind and information processing
- • Related to other fields - linguistics, computer science, philosophy, development,
- cognitive psychology Is the study of mental processes.
- •Is the study of learning, intelligence, memory, language, and thinking in humans beings.

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- •The scientific study of the human mind and information processing
- •Related to other fields -linguistics, computer science, philosophy, development, etc.
- **Components of cognitive psychology**
- Thinking
- •Intelligence
- •Language
- •Learning
- •Memory

-
- **Thinking**
-
- Thinking is a process of considering or reasoning about something.
- **terminologies.**
-
- Cognition
- •Concepts
- •Prototypes.

Cognition

- Another term for thinking, knowing and remembering
- Does the way we think really matter?

- Maybe by studying the way we think, we can eventually think better.

- **In order to think about the world, we form**

- **Concepts**

-
- A mental grouping of similar objects, events, ideas or people.
- •Concepts are similar to Piaget's idea of….
- **In order to think about the world, we form……..**

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- **Schemas**
- These animals all look different, but they fall under our concept of “dogs”.
-
-
- **We base our concepts on ….**
- **Prototypes**
-
- A mental image or best example of a category.
- We base our concepts on ….
- •If a new object is similar to our prototype, we are better able to recognize it.
- If this was my prototype of a man; then what am I?

solve problems

- **Approaches to problem solving**
- •Trial and error
- •Algorithms
- •Heuristics
- •Insight
- **trial and error**

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- also known as a mechanical solution. Trial and error refers to trying one solution after another until finding one that works.
- •For example, if Helena has forgotten the password for her email address, she can try one combination after another until she finds the one that works, if she has only a few such passwords that she normally uses.
- For example, if Helena has forgotten the password for her email address, she can try one combination after another until she finds the one that works, if she has only a few such passwords that she normally uses.

🎬 Mechanical solutions can also involve solving by rote, or a learned set of rules.

-
- •This is how word problems were solved in grade school, for example. One type of rote solution is to use an algorithm.
- **Algorithms**.
-
- Specific, step-by-step procedures for solving certain types of problems. will always result in a correct solution, if there is a correct solution to be found, and you have enough time to find it.
-
- Mathematical formulas are algorithms.
- •When librarians organize books on bookshelves, they also use an algorithm.
- But algorithms aren't always practical to use.
-
- **Heuristics**
- •A rule-of-thumb strategy that often allows us to make judgments and solve problems efficiently.
- •A short cut (that can be prone to errors).
-
- Who would you trust to baby-sit your child?
- Your answer is based on your heuristic of their appearances.
-
- **Insight**

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- •A sudden and often novel realization of the solution to a problem.
- •No real strategy involved
-
- **Four-step Process For Effective Problem Solving**
- (1) Finding and framing the problem,
- (2) Developing problem-solving strategies such as, algorithms, heuristics,
- (3) Evaluating the solutions,
- (4) Redefining the problems and solutions over time.
- **Obstacles to problem solving**
- Obstacles to problem solving
- •Confirmation Bias
- •Fixation
-
- Confirmation Bias
- •A tendency to search for information that confirms one's preconceptions.
- For example, if you believe that during a full moon there is an increase in admissions to the emergency room where you work, you will take notice of admissions during a full moon, but be inattentive to the moon when admissions occur during other nights of the month.
-
- **Fixation**
- •Mental Set
- –Match Problem
- –The Jug Problem
- –Functional Fixedness

Functional Fixedness

-
- Functional Fixedness
- •The tendency to think of things only in terms of their usual functions.
-
- Making Judgments
- •A Conditional-Reasoning Problem with a Familiar Context
-

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- Biases in making Judgment
- •Representativeness Heuristic
- •Availability Heuristic
- •Overconfidence
- •Framing
- •Belief Bias
- •Anchoring Effect
- •Belief Perseverance
-
- **Representativeness Heuristic**
- •A rule of thumb for judging the likelihood of things in terms of how well they match our prototype.
- •Can cause us to ignore important information.
- Below is Linda. She loves books and hates loud noises. Is Linda a librarian or a beautician?
- Chances are, she is a beautician!!!
- Availability Heuristic
- •Estimating the likelihood of events based on their availability in our memory.
- •If it comes to mind easily (maybe a vivid event) we presume it is common.
- Although diseases kill many more people than accidents, it has been shown that people will judge accidents and diseases to be equally fatal. This is because accidents are more dramatic and are often written up in the paper or seen on the news on T.V.
-
- **Overconfidence**
- •The tendency to be more confident than correct.
- •To overestimate the accuracy of your beliefs and judgments.
-
- **Framing**
- •The way an issued is posed.
- •It can have drastic effects on your decisions and judgments.
-
- **Belief Bias**
- •The tendency for one’s preexisting beliefs to distort logical reasoning.

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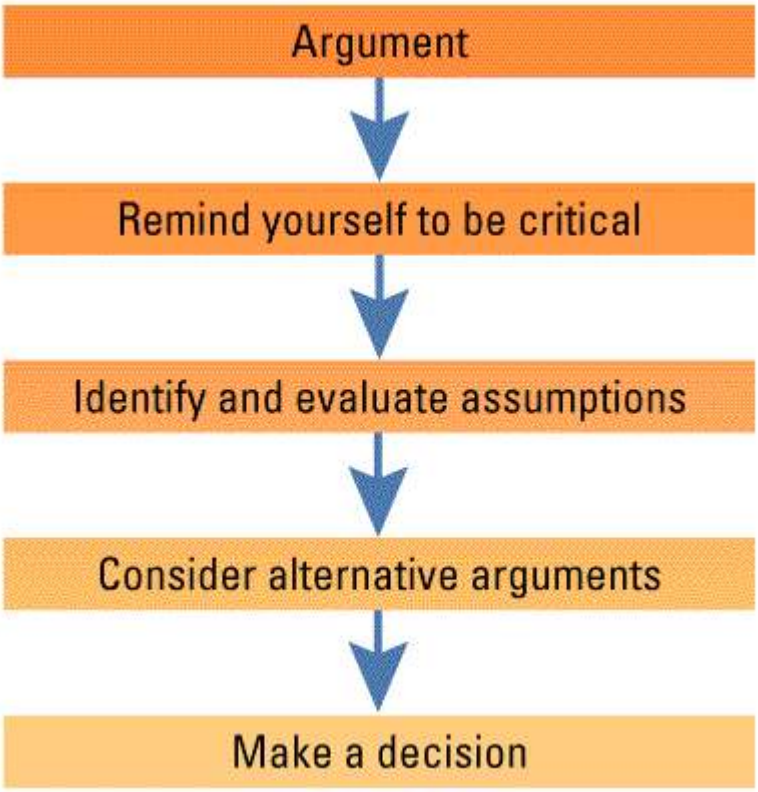
- •Sometimes making invalid conclusions valid or vice versa.
-
- **Anchoring Effect**
- The tendency to use the initial value as a reference point in making a new numerical estimate
-
- **Belief Perseverance**
- •Clinging to your initial conceptions after the basis on which they were formed has been discredited.
-
- **Critical Thinking**
- •Solving Problems and Making Judgments requires Critical Thinking.
- ♣Critical Thinking is the process of solving problems and making decisions through careful evaluation of evidence.
- **steps in critical thinking**



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- **Creativity**
- •is the ability to generate, valuable ideas.

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- •People need a minimum level of intelligence to be creative, but not all people who get high scores on intelligence tests are creative.
-
- **Divergent vs. Convergent Thinking**
- •Creativity is characterized by divergent thinking.
- •In divergent thinking, people's thoughts go off in different directions as they try to generate many different solutions to a problem.
- •In convergent thinking, on the other hand, people narrow down a list of possibilities to arrive at a single right answer.
-
- Example: Cindy would have to use divergent thinking if her professor asked her to think of a hundred different uses of a fork. She uses convergent thinking when she considers the list of possibilities on a multiple-choice question and picks the one correct answer.
-
- **Characteristics of Creative People**
- •Researchers have identified several characteristics that creative people share:
- •Expertise:Creative people usually have considerable training, knowledge, and expertise in their field.
-
- Nonconformity: Creative people tend to think independently and have relatively little concern for what others think of them.
-
- Curiosity:they tend to be open to new experiences and willing to explore unusual events.
- •Persistence: they are usually willing to work hard to overcome obstacles and take risks.
- Intrinsic motivation: they tend to be motivated more by intrinsic rewards, such as a sense of accomplishment or satisfaction of curiosity, rather than by extrinsic rewards, such as money or social approval.
-
- **Environmental Influences on Creativity**
- •People can best realize their creative potential if they are in circumstances that promote creativity. Families, organizations, and

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institutions promote creativity when they allow people to have control over problem solving and task completion,
- minimize judgement and evaluation of work, and encourage new ways of doing things.

Topic 1: Summary

- • Cognitive psychology is the scientific study of the human mind and information processing.
- • The major components of cognitive psychology are thinking, learning, intelligence, memory and language.
- • Cognition,concept and prototype are important terminologies in thinking.
- • Thinking is important in problem solving and making judgements.
- • Solving Problems and Making Judgments requires Critical Thinking.

Topic Two: Learning

-
-

Learning Objectives

- Define learning.

- ● Describe the theories of classical conditioning.
- ● Discuss the key concepts in classical conditioning
- ● Discuss the principles of classical conditioning.

Definition of learning

-

- ● A change in behavior as a result of experience or practice.

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- ● The acquisition of knowledge.
- ● Knowledge gained through study.
- ● To gain knowledge of, or skill in, something through study, teaching, instruction or experience.
- ● The process of gaining knowledgeA process by which behavior is changed, shaped or controlled.
- ● The individual process of constructing understanding based on experience from a wide range of sources
- **So, how do people learn**
- • Easy answer: We don't know for sure.
- Difficult answer: We have multiple theories that provide glimpse of an answer from many different perspectives
- These stem from psychologists, philosophers, sociologists, anthropologists, evolutionary biologists, linguists, neuroscientists
- **Broad Domains of Learning Theories**
- ● Behaviorism
- ● Cognitivism
- ● Constructivism
- ● Social learning

Behaviorism

Confined to observable and measurable behavior

- Classical Conditioning - Pavlov
- Operant Conditioning - Skinner

Classical Conditioning – Pavlov

A stimulus is presented in order to get a response**:**

S for stimuli and **R** for response

Pavlov's Conditioned Reflex

- • Conditioning -- a stimulus that initially produces no response can acquire the ability to produce one.
- • Learning occurs through pairing in time and place of one stimulus with another stimulus that produces a response
- **Conditioned Response (CR)**
- • The learned response, similar to the UCR, an involuntary reflex

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- • Pavlov distinguished several key processes that are involved in classical conditioning. These are known as:
- ❑ Acquisition,
- ❑ extinctistimulus, generalisation,
- ❑ stimulus discrimination and
- ❑ spontaneous recovery.
- **Acquisition**
- • Each paired presentation of the CS with UCS is referred to as a trial.
- • Acquisition is the overall process during which the organism learns to associate 2 events.
- • The rate of learning is often very fast in the early stages of the acquisition phase.
- • Timing of the CS and UCS pairing is critical.
- • Pavlov found that a very short time between presentations of the 2 stimuli was most effective.
- • Acquisition is more rapid when the CS occurs and remains present until the UCS is presented.
- • The end of the acquisition stage is said to occur when the CS alone produces the CR.
- **Extinction**
- • A conditioned stimulus-response association can fade over time or disappear altogether.
- • Extinction is the gradual decrease in the strength or rate of a CR that occurs when the UCS is no longer presented.
- • Extinction occurs when a CR no longer occurs following presentation of the CS.
- • E.g. Pavlov's dogs ceased salivating (CR) in response to the bell (CS) presented alone after a number of trials in which the food (UCS) did not follow the sound of the bell).
- • There is some variation between individuals in the rate at which extinction of the same conditioned response will occur.
- There is also considerable variation between the rates at which different response will be extinguished
- **Spontaneous Recovery**
- • Extinction of a CR is not always permanent.

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- • In CC, spontaneous recovery is the reappearance of a CR when the CS is presented, following a rest period after the CR appears to have been extinguished.
- • Spontaneous recovery does not always occur and when it does it is often short-lived.
- • Furthermore the CR tends to be weaker than it was originally.
-

Topic Two: Stimulus Generalization

-
-

Stimulus Generalization

- • Pavlov observed that his dogs salivated to other noises that sounded like a bell.
- • This is known as stimulus generalisation which is the tendency for another stimulus to produce a response that is similar to the CR.

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• The greater the similarity between stimuli, the greater the possibility that a generalisation will occur.

- E.g. is a stimulus generalisation to the sounds of a bell occurred with one of Pavlov's dogs, the dog might also salivate in response to the ringing of the front-door bell
-
- • However, the amount of saliva produced by the dog would tend to be less than the amount produced by the original bell to which the dog was conditioned.
- **Stimulus Discrimination**
- • Stimulus discrimination occurs when a person or animal responds to the CS only, but not to any other stimulus that is similar to the CS.

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- • E.g. in a CC experiment, stimulus discrimination would be observed when a dog salivated
- • *only* in response to the sound of the ‘experimental bell’, and not in response to any other similar sound such as a door bell
- **Classical Conditioning**
- • Behaviours that have been classically conditioned may occur so automatically that they appear to be reflexive.
- • CC behaviours are like reflexes in that they occur involuntarily, but they are unlike reflexes in that they are learned.
- • A **conditioned reflex** -an automatic response that occurs as the result of previous experience.
- • A conditioned reflex involves little conscious thought or awareness on the part of the learner. E.g listening for thunder when you see lightning
- **Conditioned emotional response**
- • An emotional reaction such as fear of a specific stimulus is learned through CC.
- • A *conditioned emotional response* is an emotional reaction that usually occurs when the autonomic nervous
- • system produces a response to a stimulus that did not previously trigger that response.
- • E.g. fearing the sound of the dentist’s drill.

Topic Two: Watson’s little Albert’s experiment

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Watson’s little Albert’s experiment

- • American psychologist John B. Watson and his graduate student, Rosalie Rayner first used CC to elicit an emotional response.
- • Aim to test the notion that fears can be acquired through.

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- How was Little Albert conditioned to hate the rat?
- • They placed him on a mattress in a room where a white lab rat (CS) was within reaching distance.
- • Albert showed no initial fear of it and played with it.
- • They then struck a hammer on a steel bar behind Albert (loud noise, UCS) and Albert began to cry. For the next 17 days Watson and Rayner began a series of fear-conditioning experiments.
- • They also conducted tests to find out if Albert's fear response could be generalised
- • Albert also seemed to fear a white rabbit, a dog and a seal skin coat

Topic two: Summary

- • Learning is a process by which behaviour is changed, shaped or controlled.
- • The broad domains of learning are Behaviourism, Cognitivism, Constructivism and social learning.
- • Behaviourism is confined to observable and measurable behaviour i.e Classical Conditioning and Operant Conditioning.
- • Positive reinforcement is the presentation of a stimulus after a response so that the response will occur more often.

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- • Negative reinforcement is the removal of an a aversive stimulus after a response so that the response will occur more often.
- • Positive punishment is the presentation of a stimulus after a response so that the response will occur less often.
- Negative punishment is the removal of a stimulus after a response so that the response will occur less often.

Topic Three: Intelligence.

-
-

- **INTELLIGENCE**
- **Objectives**
- 1. Define intelligence.
- 2. Describe the theories of intelligence.
- 3. Describe how intelligence is measured.
- 4. Discuss the factors influencing intelligence.
- **Intelligence**
- The capacity to understand the world, think rationally, and use resources effectively when faced with challenges
-

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- ❖ To psychologists, intelligence is the capacity to understand the world, think rationally, and use resources effectively when faced with challenges (Wechsler, 1975).
- ❖ Psychologists who study intelligence have focused much of their attention on the development of intelligence tests, and have relied on such tests to identify a person's level of intelligence. These tests have proven to be of great benefit in identifying students in need of special attention in school, in diagnosing cognitive difficulties, and in helping people make optimal educational and vocational choices. At the same time, their use has proven quite controversial, raising important social and educational issues.
- Measuring Intelligence

- ■ Mental age
- – The average age of individuals who achieve a particular level of performance on a test
- ■ Chronological age
- – Physical ageThe forerunner of the modern IQ test was based on an uncomplicated, but completely wrong, assumption: that the size and shape of a person's head could be used as an objective measure of intelligence. The idea was put forward by Sir Francis Galton (1822-1911), an eminent English scientist whose ideas in other domains proved to be considerably better than his notions about intelligence.
- – Galton's motivation to identify people of high intelligence stemmed from his prejudices. He sought to demonstrate the natural superiority of people of high social class (of which he was one) by showing that intelligence was inherited. He hypothesized that head configuration, being genetically determined, was related to brain size, and therefore related to intelligence.
- – Galton's theories proved wrong on virtually every count. Head size and shape were not related to intellectual performance, and subsequent research has found little relationship between brain size and intelligence. However, Galton's work did have at least one desirable result: He was the

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first person to suggest that intelligence could be quantified and measured in an objective manner.

- – Following Galton's efforts, the first intelligence tests were developed by Alfred Binet (1857-1911). His tests followed a simple premise: If performance on certain tasks or test items improved with *chronological*, or physical, age, then performance could be used to distinguish more intelligent people from less intelligent ones within a particular age group. Using this principle, Binet, a French psychologist, devised the first formal intelligence test, which was designed to identify the "dullest" students in the Paris school system in order to provide them with remedial aid.
- ❖ On the basis of the Binet test, children were assigned a score relating to their mental age, the average age of individuals who achieve a particular level of performance on a test. For example, if the average 8-year-old answered, say, 45 items correct on a test, anyone who answered 45 items correct would be assigned a mental age of 8 years. Consequently, whether the person taking the test was 20 years old or 5 years old, each would have the same mental age of 8 years.
- ❖ Assigning a mental age to students provided an indication of their general level of performance. However, it did not allow for adequate comparisons among people of different chronological ages. By using mental age alone, for instance, we might assume that a 20-year-old responding at a 18-year-old's level would be as bright as a 5-year-old answering at a 3-year-old's level, when actually the 5-year-old would be displaying a much greater *relative* degree of slowness.
- ❖ A solution to the problem came in the form of the intelligence quotient, or IQ, a score that takes into account an individual's mental *and* chronological ages. To calculate an IQ score, the following formula is used, in which MA stands for mental age and CA for chronological age:
- ❖ Although the basic principles behind the calculation of an IQ score still hold, IQ scores actually are figured in a different manner today and are known as *deviation IQ scores*. First, the average test score for everyone of the same age who takes the test is determined, and this average score is assigned an IQ of

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100. Then, with the aid of statistical techniques that calculate the differences (or "deviations") between each score and the average, IQ scores are assigned.

- **Gauging Intelligence**

- ❖ Remnants of Binet's original intelligence test are still with us, although it has been revised in significant ways. Now in its fourth edition and called the *Stanford-Binet IV*, the test consists of a series of items that vary in nature according to the age of the person being tested (Hagen, Sattler, & Thorndike, 1985; Thorndike, Hagan, & Sattler, 1986). For example, young children are asked to copy figures or answer questions about everyday activities. Older people are asked to solve analogies, explain proverbs, and describe similarities that underlie sets of words.

- ❖ By examining the pattern of correct and incorrect responses, the examiner is able to compute an IQ score for the person being tested. In addition, the Stanford Binet provides separate subscores, providing clues to a test-taker's particular strengths and weaknesses.

- ❖ The IQ test most frequently used in the United States was devised by psychologist David Wechsler and is known as the *Wechsler Adult Intelligence Scale—III*, or, more commonly, the *WAIS-III*. There is also a children's version, the *Wechsler Intelligence Scale for Children—III*, or *WISC-III*. Both the WAIS-III and the WISC-III have two major parts: a verbal scale and a performance (or nonverbal) scale.The two scales include questions of very different types. Verbal tasks consist of more traditional kinds of problems, including vocabulary definition and comprehension of various concepts. In contrast, the performance (nonverbal) part involves the timed assembly of small objects and arranging pictures in a logical order. Although an individual's scores on the verbal and performance sections of the test are generally within close range of each other, the scores of a person with a language deficiency or a background of severe environmental deprivation may show a relatively large discrepancy between the two scores. By providing separate scores, the WAIS-III and WISC-III give a more precise picture of a person's specific abilities than other IQ tests (Kaufman & Lichtenberger, 1999).

- **Achievement and Aptitude Tests**

- ❖ An **achievement test** is a test designed to determine a person's level of knowledge in a given subject area. Rather than measuring general ability, as an intelligence test does, an achievement test concentrates on the specific material that a person has learned. High school students sometimes take specialized achievement tests in particular areas such as world history or chemistry as a college entrance requirement; lawyers must pass an achievement test (in the form of the bar exam) in order to practice law.

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- ❖ An **aptitude test** is designed to predict a person's ability in a particular area or line of work. Most of us take one of the best-known aptitude tests in the process of pursuing admission to college: the SAT(Scholastic Achievement Test). The SAT is meant to predict how well people will do in college, and the scores have proven over the years to be moderately correlated with college grades.

- **Reliability and Validity**

- ❖ In the same way, we hope that psychological tests have **reliability**—that they measure consistently what they are trying to measure. We need to be sure that each time we administer the test, a test-taker will achieve the same results—assuming that nothing about the person has changed relevant to what is being measured. A test has **validity** when it actually measures what it is supposed to measure. Knowing that a test is reliable is no guarantee that it is also valid. For instance, Sir Francis Galton assumed that skull size was related to intelligence, and he was able to measure skull size with great reliability. However, the measure of skull size was not valid—it had nothing to do with intelligence. In this case, then, we have reliability without validity. In Canada, the Nelson-Denny reading test and a working memory test were the best predictors of SAT scores (Daneman and Hanon, 2001).

- ❖ On the other hand, if a test is unreliable, it cannot be valid.

- ❖ Test validity and reliability are prerequisites for accurate assessment of intelligence—as well as for any other measurement task carried out by psychologists.

- ❖ Assuming that a test is both valid and reliable, one further step is necessary in order to interpret the meaning of a particular test-taker's score: the establishment of norms. **Norms** are standards of test performance that permit the comparison of one person's score on the test to the scores of others who have taken the same test. For example, a norm permits test-takers to know that they have scored, say, in the top 15 percent of those who have taken the test previously. Tests for which norms have been developed are known as *standardized tests.*

- ❖ There is now a computerized version of the SAT. Using *adaptive testing*, the computer will generate easier or more difficult questions depending on the test-takers answers on the previous questions.

- ❖ One central question addressed by researchers is whether intelligence is a single, unitary factor, or whether it is made up of multiple components. Early psychologists interested in intelligence assumed that there was a single, general factor for mental ability, which they called **g**, or **g-factor** (Spearman, 1927). This general intelligence factor was thought to underlie performance on every aspect of intelligence, and it was the g-factor that was presumably being measured on tests of intelligence (Jensen, 1998; Mackintosh, 1998).

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- ❖ Fluid and crystallized intelligence. More recently, some psychologists have suggested that there are really two different kinds of intelligence: fluid intelligence and crystallized intelligence (Cattell, 1967, 1987). **Fluid intelligence** reflects information processing capabilities, reasoning, and memory. If we were asked to solve an analogy, group a series of letters according to some criterion, or to remember a set of numbers, we would be using fluid intelligence.

- ❖ In contrast, **crystallized intelligence** is the accumulation of information, skills, and strategies that people have learned through experience and that they can apply in problem-solving situations. We would be likely to rely on crystallized intelligence, for instance, if we were asked to participate in a discussion about the solution to the causes of poverty, a task that allows us to draw upon our own past experiences and knowledge of the world. The differences between fluid and crystallized intelligence become particularly evident in the elderly, who—as we will discuss further in Chapter 10—show declines in fluid, but not crystallized, intelligence (Schaie, 1993, 1994; Schretlen, 2000).

- ❖ In his consideration of intelligence, psychologist Howard Gardner has taken a very different approach from traditional thinking about the topic. Gardner argues that rather than asking "How smart are you," we should be asking a different question: "How are you smart?" In answering the latter question, Gardner has developed a theory of *multiple intelligences* that has become quite influential (Chen & Gardner, 1997).

- ❖ Gardner argues that we have eight different forms of intelligence (described in Figure 7-11), each relatively independent of the others. He suggests that each of the multiple intelligences is linked to an independent system in the brain (Gardner, 1997, 1999).

- ❖ Although Gardner illustrates his conception of the specific types of intelligence with descriptions of well-known people, each of us has the same eight kinds of intelligence, although in different degrees. Moreover, although the eight are presented individually, Gardner suggests that these separate intelligences do not operate in isolation. Normally, any activity encompasses several kinds of intelligence working together.

- Gardner's theory has led to the development of intelligence tests that include questions in which more than one answer can be correct, providing the opportunity for test takers to demonstrate creative thinking. These tests are based on the idea that different kinds of intelligence may produce different—but equally valid—responses to the same question.

- **Information Processing Intelligence?**

- ■ Cognitive psychologists believe intelligence is indicated by how information is processed:

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- – The processes used to produce intelligent behaviour
- • Encoding, identifying parts of problems, etc
- – The speed of processing
- ■ **Practical intelligence** is intelligence related to overall success in living
- ■ Noting that traditional tests were designed to relate to academic success, Sternberg points to evidence showing that IQ does not relate particularly well to *career* success
- ■ Sternberg argues that career success requires a very different type of intelligence from academic success. Whereas academic success is based on knowledge of a particular information base obtained from reading and listening, practical intelligence is learned mainly through observation of others' behaviour. People who are high in practical intelligence are able to learn general norms and principles and apply them appropriately. See Figure 7-12.
- ■ **Emotional intelligence** is the set of skills that underlie the accurate assessment, evaluation, expression, and regulation of emotions (Goleman, 1995; Mayer & Salovey, 1997; Salovey & Sluyter, 1997).
- ■ According to psychologist Daniel Goleman (1995), emotional intelligence underlies the ability to get along well with others. It provides us with the understanding of what other people are feeling and experiencing, and permits us to respond appropriately to others' needs. Emotional intelligence is the basis of empathy for others, self-awareness, and social skills.
- ■ **Canadian Steven** Stein distinguishes intrapersonal aspects (being emotionally self-aware) of emotional intelligence from interpersonal aspects (recognizing emotions in others (Stein & Book, 2000)).
- ■ Abilities in emotional intelligence may help explain why people with only modest IQ scores can be quite successful, despite their lack of traditional intelligence. High emotional intelligence may enable an individual to tune into others' feelings, permitting a high degree of responsiveness to others.
-
- **Variations in Intellectual Ability**
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- ❖ Although sometimes thought of as a rare phenomenon, mental retardation occurs in 1 to 3 percent of the population. There is wide variation among those labeled as mentally retarded, in large part because of the inclusiveness of the definition developed by the American Association on Mental Retardation. The association suggests that **mental retardation** exists when there is significantly below-average intellectual functioning, plus limitations in at least two areas of adaptive functioning involving communication skills, self-care, ability to live independently, social skills, community involvement, self direction, health and safety, academics, or leisure and work (AAMR, 1992; Burack, Hodapp, & Zigler, 1998).
- ❖ In Canada, mental retardation is now referred to as *intellectual disability or developmental disability.*
- ❖ *Intellectual disability* refers to who have mild deficits (poor daily living and social skills, but functional reading and mathematics skills). These are most biological in origin (*Down Syndrome, FAS,* lack of oxygen at birth)
- ❖ *Developmental disability* refers to those who have profound disability (moderate to severe intellectual deficits as well as physical challenges.
- ❖ While below average intellectual functioning can be measured in a relatively straightforward manner—using standard IQ tests—it is more difficult to determine how to gauge limitations in particular adaptive skill areas. Ultimately, this imprecision leads to a lack of uniformity in how experts apply the label of "mental retardation." Furthermore, it has resulted in significant variation in the abilities of people who are categorized as mentally retarded, ranging from those who can be taught to work and function with little special attention, to those who virtually cannot be trained and must receive institutional treatment throughout their lives (Durkin & Stein, 1996; Negrin & Capute, 1996; Detterman, Gabriel, & Ruthsatz, 2000).
- ▪ Communication skills
- ▪ Self-care
- ▪ Ability to live independently
- ▪ Social skills
- ▪ Community involvement

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- ▪ Self direction
- ▪ Health & safety
- ▪ Academics
- ▪ Leisure & work

Identifying Roots of Mental Retardation

- ❖ In most of Canada and the USA, legislation supports mainstreaming which is the full integration of individuals with intellectual or developmental disabilities into regular classrooms. This was in part to remove the isolation and stigma experienced in special education classes.

- ❖ The most common biological cause of retardation is Down syndrome. *Down syndrome* occurs due to the presence of an extra chromosome. In other cases of mental retardation, an abnormality occurs in the structure of a chromosome. Birth complications, such as a temporary lack of oxygen, may also cause retardation (Simonoff, Bolton, & Rutter, 1996; Selikowitz, 1997).

- ❖ The majority of cases of mental retardation are classified as *familial retardation*, in which no known biological defect exists, but there is a history of retardation within the family. Whether the family background of retardation is caused by environmental factors, such as extreme continuous poverty leading to malnutrition, or by some underlying genetic factor, is usually impossible to determine.

- ❖ Regardless of the cause of mental retardation, important advances in the care and treatment of those with retardation have been made in the last several decades. In this USA federal law, Congress ruled that people with retardation are entitled to a full education and that they must be educated and trained in the *least restrictive environment*. The law increased the educational opportunities for individuals with mental retardation, facilitating their integration into regular classrooms as much as possible—a process known as *mainstreaming* (Hocutt, 1996; Lloyd, Kameenui, & Chard, 1997).

- Mandatory legislation was in place in Saskatchewan in 1971, Nova Scotia in 1973 and in Ontario in 1980. This legislation supported mainstreaming which is the full integration of individuals with intellectual or developmental disabilities into regular classrooms. This was in part to remove the isolation and stigma experienced in special education classes. Regular classrooms will improve their educational opportunities, increase their social acceptance, and facilitate their integration into society.

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- Intellectually Gifted

- ❖ Composing 2 to 4 percent of the population, the intellectually gifted have IQ scores greater than 130.

- ❖ Although the stereotype associated with the gifted suggests that they are awkward, shy, social misfits unable to get along well with peers, most research indicates that just the opposite is true. The intellectually gifted are most often outgoing, well-adjusted, popular people who are able to do most things better than the average person

- ❖ For example, in a long-term study by psychologist Lewis Terman that started in the early 1920s and is still going on, 1500 children who had IQ scores above 140 were followed and examined periodically over the next sixty years (Terman & Oden, 1947; Sears, 1977). From the start, members of this group were more physically, academically, and socially capable than their nongifted peers. They were generally healthier, taller, heavier, and stronger than average. Not surprisingly, they did better in school as well. They also showed better social adjustment than average. And all these advantages paid off in terms of career success: As a group, the gifted received more awards and distinctions, earned higher incomes, and made more contributions in art and literature than typical individuals.

- ❖ Of course, not every member of the group Terman studied was successful. Furthermore, high intelligence is not a homogeneous quality; a person with a high overall IQ is not necessarily gifted in every academic subject, but may excel in just one or two. A high IQ, then, is no universal guarantee of success (Shurkin, 1992; Winner, 2000).

- **Individual Differences in Intelligence**

- ❖ In an attempt to produce a **culture-fair IQ test**, one that does not discriminate against members of any minority group, psychologists have tried to devise test items that assess experiences common to all cultures or emphasize questions that do not require language usage. However, test makers have found

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this difficult to do, because past experiences, attitudes, and values almost always have an impact on respondents' answers.

- ❖ Members of certain racial and cultural groups consistently score lower on traditional intelligence test than members of other groups (MacKenzie, 1984; Humphreys, 1992). Blacks tend to average 10-15 IQ points lower than whites. To combat this with the Cree of James Bay, Mawhinney and collaborators from the Cree people developed the Cree Picture Vocabulary Test (Mawhinney, 1983).

- ❖ The efforts of psychologists to produce culture-fair measures of intelligence relate to a lingering controversy over differences in intelligence between members of minority and majority groups. In attempting to identify whether there are differences between such groups, psychologists have had to confront the broader issue of determining the relative contribution to intelligence of genetic factors (heredity) and experience (environment)—the nature-nurture issue that we first discussed in Chapter 1

- ❖ Richard Herrnstein, a psychologist, and Charles Murray, a sociologist, fanned the flames of the debate with the publication of their book, *The Bell Curve,* in the mid-1990s. They argued that an analysis of IQ differences between whites and blacks demonstrated that, although environmental factors played a role, there were also basic genetic differences between the two races.

- ❖ Figure 7-13 indicates that Intelligence in general shows a high degree of heritability, a measure of the degree to which a characteristic is related to genetic, inherited factors.

- ❖ However, many psychologists reacted strongly to the arguments laid out in *The Bell Curve,* refuting several of the book's contentions. For one thing, even when attempts are made to hold socioeconomic conditions constant, wide variations remain among individual households. Furthermore, no one can convincingly assert that living conditions of blacks and whites are identical even when their socioeconomic status is similar.

- ❖ It is also crucial to remember that IQ scores and intelligence have greatest relevance in terms of individuals, not groups. In fact, considering group *racial* differences presents some conceptually troublesome distinctions. Consequently, drawing comparisons between different races on any dimension, including IQ scores, is an imprecise, potentially misleading, and often fruitless venture. By far, the greatest discrepancies in IQ scores occur when comparing *individuals*, and not when comparing mean IQ scores of different *groups*.

- ❖ Other issues make the heredity-versus-environment debate somewhat irrelevant to practical concerns. For example, as we discussed earlier, there are multiple kinds of intelligence, and traditional IQ scores do not tap many of them.

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Furthermore, IQ scores are often inadequate predictors of ultimate occupational success.

- ❖ IQs have been steadily improving since 1920s. This may be due to better nutrition, better parenting, improved education, improved social environment, increased skills related to technological changes or some combination of these factors.

Topic four: Memory

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Learning objectives

- ● Describe the stages of processing memory
- ● Describe the stages of memory
- ● Identify the two forms of memory
- ● Describe two types of memory processing.
- ● Explain the factors that influence forgetfulness.
-
- **Memory** is an active system that stores, organizes, alters, and recovers (retrieves) information.

- **Stages of processing Information**
- ● Memory processes mental representations in a series of stages.
- ● Encoding, storage, and retrieval are stages of processing in memory, for example.
- **Stages of memory**
- 1. Sensory Memory
- 2. Short-Term Memory
- 3. Long-Term Memory
- **Graphic Representation on Stages of Memory**

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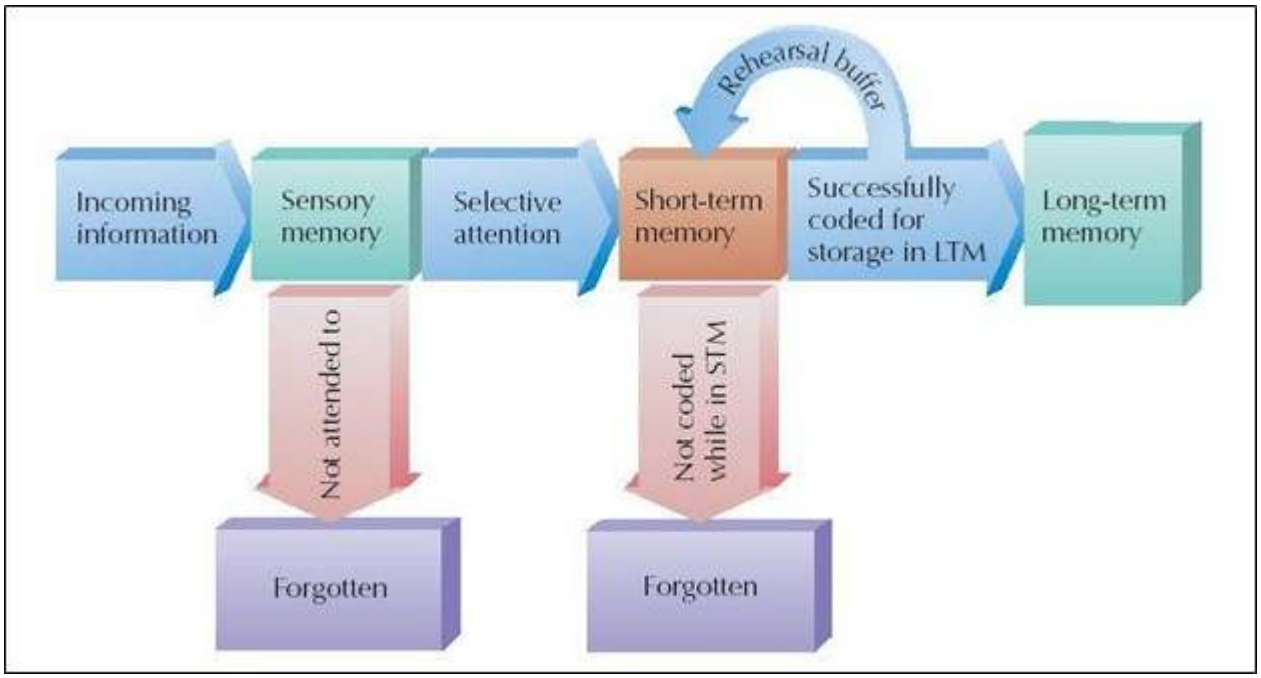

- **Sensory Memory**
- • Sensory Memory: Storing an exact copy of incoming information from the sensory system for less than a second; the first stage of memory. It is recorded by the person's sensory system as a raw, non meaningful stimulus. Holds large amount of perceptual input for a very brief period of time.
- • Iconic: The process that reflects information from our visual system. Last less than a second (Long & Beaton, 1982)
- • Echoic: The process that stores information coming from the ears. Fades within 3 or 4 seconds (Darwin, Turvey & Crowder, 1972)
- **Short-Term Memory**
- • **Short-Term Memory (STM):** second stage of memory; stores small amounts of information briefly; very sensitive to interruption or interference
- • **Phonetically**: Storing information by *sound*; how most things are stored in STM
- • **Memory capacity:** STM is limited to holding seven (plus or minus two) information bits at once.

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- • **Chunk**: To organize meaningful units of information in memory. Increases recall capacity
- **Storing Info in STM**
- • **Recoding**: Reorganizing or modifying information in STM
- • **Maintenance Rehearsal**: Repeating information silently to prolong its presence in STM
- • **Elaborative Rehearsal**: Links new information with existing memories and knowledge in LTM; Good way to transfer STM information into LTM
- **Long-Term Memory (LTM)**
- • Storing information relatively permanently
- • Stored on basis of meaning and importance
- **Types of Long-Term Memory**
- • **Declarative memory (facts)**: factual knowledge & personal experiences
- ❑ **Semantic Memory**: Memory for general knowledge an facts about the world
- ❑ **Episodic Memory**: Personal experiences linked with specific times and places
- • **Procedural Memory (skills)**: Long-term memories of conditioned responses and learned skills, e.g., driving
- Two forms of Memory

- • Explicit memory – refers to intentional or conscious recollection of information.
- • Implicit memory – refers to memories of which people are not consciously aware, but which affect subsequent performance and behavior.
-
- **Types of Memory Processing**
- • **Automatic processing:** memory processing that occurs subconsciously and does not require attention.
- *Example: How many of you can sing the National Anthem? How many learned it on purpose?*
- • **Effortful processing:** memory processing that occurs consciously and requires attention

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- *Example: How many of you can name all of the divisions of the nervous system? How many learned it on purpose*
- **Forgetting**
- • The process of forgetting in memory storage refers to our inability to retrieve a piece of information that has previously been stored.
- • Forgetting is different to amnesia because it only relates to missing pieces of memory rather than the entire episode..
- **Why we forget?**
- • Decay
- • Replacement
- • Interference
- • Cue-dependent forgetting
- • Psychogenic Amnesia
- **Decay**
- • Decay theory – the theory that information in memory eventually disappears if it is not accessed; it applies more to short –term than to long term.
- • Rehearsal does not strengthen items in short term memory but merely prevents them from gradually disappearing
- **Interference**
- • **Retroactive interference** – forgetting that occurs when recently learned material interferes with the ability to remember similar material stored previously.
- • **Proactive interference** – forgetting that occurs when previously stored material interferes with the ability to remember similar, more recently learned material.
- **Cue- dependent forget**
- • The inability to retrieve information stored in memory because of insufficient cues for recall.
- ❑ State dependent memory: the tendency to remember something when the rememberer is in the same physical or mental state as during the original learning
- **Psychogenic amnesia or Experience**
- • The partial or complete loss of memory (due to nonorganic causes) for threatening information or traumatic experiences.

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- • **Childhood amnesia** (the missing years) – the inability to remember events and experiences that occurred during the first two or three years of life.
- • **Anterograde amnesia:** the inability to form new memories for events following brain trauma or surgery. Explicit memories formed before are left intact. Cause possibly by damage to hippocampus (medial temporal lobe)
- • **Retrograde amnesia:** the disruption of memory for the past, especially espisodic memory. After brain trauma or surgery, there often is retrograde amnesia for events occurring just before.
- **Improving your Memory**
- • The keyword technique
- • Method of loci
- • The encoding specificity phenomenon
- • Organization of text material
- • Organization of lecture notes
- • Practice and rehearsal
- **LANGUAGE**

- ❖ Our ability to make sense out of nonsense, if the nonsense follows typical rules of language, illustrates both the sophistication of human language capabilities and the complexity of the cognitive processes that underlie the development and use of language. The use of **language—**the communication of information through symbols arranged according to systematic rules—clearly represents an important cognitive ability, one that is indispensable for communicating with others. Not only is language central to communication, it is also closely tied to the very way in which we think about and understand the world, for there is a crucial link between thought and language. It is not surprising, then, that psychologists have devoted considerable attention to studying the topic of language (Forrester, 1996; Velichkovsky & Rumbaugh, 1996; Barrett, 1999; Owens, 2000).

- ❖ **Syntax** refers to the rules that indicate how words and phrases can be combined to form sentences. Every language has intricate rules that guide the order in which words may be strung together to communicate meaning. English speakers have no difficulty recognizing that “Radio down the turn” is not an appropriate sequence, while “Turn down the radio” is. The importance of appropriate syntax is demonstrated in English by the changes in meaning that are caused by the different word orders in the following three utterances: “John kidnapped the boy,” “John, the kidnapped boy,” and “The boy kidnapped John” (Lasnik, 1990).

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- ❖ The third major component of language is semantics. **Semantics** refers to the rules governing the meaning of words and sentences (Larson, 1990; Hipkiss, 1995; O'Grady & Dobrovolsky, 1996). Semantic rules allow us to use words to convey the subtlest of nuances. For instance, we are able to make the distinction between "The truck hit Laura" Children **babble**—make speechlike but meaningless sounds—from around the age of 3 months through 1 year. While they babble they may produce, at one time or another, any of the sounds found in all languages, not just the one to which they are exposed.

- ❖ Babbling increasingly reflects the specific language that is being spoken in an infant's environment, initially in terms of pitch and tone, and eventually in terms of specific sounds. Some theorists suggest there is a *critical period* for language development early in life, in which a child is particularly sensitive to language cues and during which language is most easily acquired.

- ❖ By the time the child is approximately 1 year old, sounds that are not in the language to which the infant is exposed disappear. It is then a short step to the production of actual words. In English, these are typically short words that start with a consonant such as "b," "d," "m," "p," or "t". Of course, even before they produce their first words, children are capable of understanding a fair amount of the language they hear.

- ❖ After the age of 1 year, children begin to learn more complicated forms of language. They produce two-word combinations, which become the building blocks of sentences, and the number of different words they are capable of using increases sharply. By the age of 2 years, the average child has a vocabulary of more than fifty words. Just six months later, that vocabulary has grown to several hundred words. At that time, children can produce short sentences, although they use **telegraphic speech**—sentences that sound as if they were part of a telegram, in which words not critical to the message are left out. Rather than saying, "I showed you the book," a child using telegraphic speech might say, "I show book"

By age 3, children learn to make plurals by adding "s" to nouns and to form the past tense by adding "ed" to verbs. This ability also leads to errors, since children tend to apply rules too inflexibly. This phenomenon is known as **overgeneralization**, whereby children apply rules even when the application results in an error. Thus, although it is correct to say "he walked" for the past tense of "walk," the "ed" rule doesn't work quite so well when children say "he runned" for the past tense of "run" (Marcus, 1996

Understanding Language Acquisition

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❖ Noam Chomsky (1968, 1978, 1991), a linguist argued that humans are born with an innate linguistic capability that emerges primarily as a function of maturation. According to his analysis, all the world's languages even for deaf children, share a similar underlying structure. Chomsky suggests that the human brain has a neural system, the **language-acquisition device**, that both permits the understanding of the structure of language and provides strategies and techniques for learning the unique characteristics of a given native language.

❖ The **learning-theory approach** suggests that language acquisition follows the principles of reinforcement and conditioning discussed in Chapter 5. For example, a child who utters the word "mama" is hugged and praised by her mother, which reinforces the behaviour and makes its repetition more likely. This view suggests that children first learn to speak by being rewarded for making sounds that approximate speech. Ultimately, through a process of shaping, language becomes more and more like adult speech (Skinner, 1957).

❖ Canada has two official languages but there are many languages spoken by Canadians. Bilinguals have distinct advantages compared to monolinguals.
See *Exploring Diversity* and *Pathways Through Psychology sections.*

❖ The learning theory approach is less successful when it comes to explaining the acquisition of language rules. Children are reinforced not only when they use proper language, but also when they respond incorrectly.

Canadian teachers have used the whole language approach, the basic skills approach, or a combination of the two.

❖ The *whole language approach* stresses the use of whole and meaningful materials. This 'top-down' processing means that children develop a story orally. Then the teacher prints the story and they read it.

❖ The *basic skills approach* emphasizes the recognition of words in isolation. This 'bottom-up' processing means that the children learn the rules, the sound units into words and then the words into sentences.

❖ Research by Betty Ann Levy, McMaster University indicates both basic skills and whole language approaches contribute to reading acquisition.

❖ Levy and Lysynchuk (1997) instructed one group of beginning readers in the acquisition of an initial word set using whole language approach. They instructed a second group using the basic skills approach. They found that word set was learned more quickly and accurately by readers instructed by the basic skills approach than by the whole language approach. Also, they found that the thoroughness by which the children learned the word set determined how well they read new words. Single word practice enabled the children to read stories with these words more quickly and with better comprehension.

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